



OIL INDUSTRY SAFETY DIRECTORATE

सीमेंटिंग प्रचालन ओआईएसडी - मानक - 175 CEMENTING OPERATIONS OISD - STANDARD - 175

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PREFACE

Indian petroleum industry is the energy lifeline of the nation and its continuous performance is essential for prosperity of the country. As the industry deals with inherently inflammable substances throughout its value chain – upstream, midstream and downstream – Safety is of paramount importance to this industry as safe performance at all times can ensure optimum return of these national assets and resources including sustainability.

To ensure proper implementation of the various aspects of safety in the oil & gas industry, the Government of India, vide Resolution No. R-13013/4/84-OR-1 dated 10th January, 1986 set up a Safety Council at the apex, under the administrative control of the Ministry of Petroleum & Natural Gas (Mo&PNG). To assist the Safety Council, a technical directorate, namely the Oil Industry Safety Directorate (OISD) under the aegis of MoP&NG, has been entrusted with the responsibility of formulating standards, overseeing its implementation through safety audits to enhance safety level in the oil & gas industry.

OISD has developed a rigorous, multi-layer, iterative and participative approach for development of standards. OISD develops and revises Standards, Guidelines & Recommended Practices for the oil and gas sector through a participative process involving all the stakeholders, drawing inputs from national and international standards and adapting them by leveraging the experience of the functional committee. These standards cover inbuilt design safety, asset integrity and best operating practices in the field of production, processing, storage and transport of petroleum.

The participative process followed in standard formulation has resulted in excellent level of compliance by the industry. It also goes to prove the old adage that self-regulation is the best regulation. The quality and relevance of OISD standards had been further endorsed by their adoption/ reference in various statutory rules and regulations of the land.

OISD and industry together strive to achieve nil incident in the entire hydrocarbon value chain. We at OISD, are confident that the provisions of this standard, when implemented in totality, would go a long way in ensuring safe operation in the oil and gas industry.

**Executive Director
Oil Industry Safety Directorate**

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Wherever Acts/ Rules/ Regulation and National/ International Standards are mentioned in the standard, the same relates to in-vogue version of such documents. Whenever there is any difference between the statutory norm and OISD standard, more stringent requirement should be followed. However, in case of any difference in the requirement between OISD standards, the requirement as specified in the standard released at a later stage shall prevail.

The industry has to carry out gap analysis and make necessary changes in their system in line with the revised standard and the requirement should be implemented within a time frame of 1 year (from the date of release of revised standard).

Though every effort has been made to assure the accuracy and reliability of the information contained in these documents, OISD hereby expressly disclaims any liability or responsibility for loss or damage resulting from the use of OISD standard.

The standards would be reviewed periodically based on the decision in the Steering Committee/ Safety Council, industry request, technological development, recommendations of high-level committee and accident investigation reports. Generally, OISD standards are reviewed and revised, reaffirmed or withdrawn at least every ten years.

Suggestions for revision are invited from the oil and gas industry and should be submitted to OISD at ps-ed.oisd@gov.in.

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OISD sincerely acknowledges the contributions made by industry veterans by providing his technical & professional guidance for revision of this standard. OISD also recognizes the support & cooperation of several other experts from industry, who contributed for preparation, review and finalization of this document

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**CEMENTING OPERATIONS****1.0 INTRODUCTION**

Well cementation is a vital operation in oil and gas drilling, playing a key role in maintaining well integrity. Cementation prevents formation fluid migration between subsurface formations and from ground water reservoirs into the borewell. It reinforces the casing in the well bore and safeguards it against corrosion. By providing structural stability and zonal isolation, good well cementation enhances the life of the well and efficiency of oil and gas production while minimizing potential hazards.

The cement slurry should exhibit certain hydraulic properties for proper placement in the wellbore while the set cement sheath must possess certain mechanical properties for long term structural support and zonal isolation. The set cement acts as a barrier to the undesired flow of fluids/gas from formation to other formation or to surface in a successfully drilled well.

Factors which are known to contribute to success in primary and secondary cementing operations are described to facilitate proper planning and execution. Cementation of deep-water, high angle, HPHT and CBM wells has also been briefly included.

2.0 SCOPE

This document provides operating standards and recommended practices for cementation in onshore and offshore wells for well integrity and safe operations.

3.0 DEFINITIONS

Barrier: A component or practice that, if properly installed, contributes to the total system reliability by preventing liquid or gas flow.

Bottomhole circulating temperature (BHCT): Pseudo-stable temperature found at the bottom of the wellbore while the well is being circulated.

Casing Integrity: The ability of the casing to withstand the various forces and pressures encountered during drilling and production operations.

Casing stand-off ratio: Smallest distance between the well bore and out diameter of casing expressed as percentage of annular clearance available for perfectly centred casing.

Equivalent circulating density (ECD): Equivalent circulating density is the effective density of the circulating fluid in the wellbore resulting from the sum of the hydrostatic pressure imposed by the static fluid column and the friction pressure.

HPHT well: Well in which the undisturbed bottom hole temperature at prospective reservoir depth is greater than 300°F and the maximum anticipated pore pressure of any porous formation to be drilled through exceeds - around 10000psi.

Leak-off test (LOT): A leak-off test is a procedure used to determine the wellbore pressure required to initiate a fracture in the open or exposed formations.

**CEMENTING OPERATIONS**

Lost circulation / returns: The reduced or total absence of fluid returning to the surface after being pumped down a well. Lost returns occur when the drill bit encounters natural fissures, fractures or caverns, and the fluid flows into the newly available spaces rather than returning to surface.

May: The term 'may' indicate desirable or good practices which are encouraged to be implemented.

Mechanical barrier: A subset of physical barriers that features mechanical equipment; not set cement or a hydrostatic fluid column.

Free fluid: Coloured or colourless liquid that has separated from a cement slurry when standing in a static condition.

Gas Tight Slurry: A property of cement slurry that generates gel structure rapidly after the pumping is complete, to hold formation gas or liquid from entering the annulus filled with cement slurry.

Micro Annulus: A small gap most commonly formed between the casing or liner and the surrounding cement sheath, by variations in temperature or pressure or stresses during or after the cementing process.

Permanent abandonment: The well status where the well or part of the well, will be plugged and abandoned permanently with the intention of never being used or re-entered again.

Shall: The term "shall" indicate the mandatory requirement.

Should: The term "Should" indicates that the provision is recommended as a good engineering practice.

Sustained casing pressure (SCP): Pressure in an annulus of casing strings that is measurable at the wellhead that rebuilds to at least the same pressure level after pressure has been bled down. SCP is not due solely to temperature induced fluid expansion or a pressure that has been imposed by the operator. See API RP 90 for further information.

Thickening time: Time elapsed from the initial application of pressure and temperature to the time at which the slurry reaches a consistency deemed sufficient to make it unpumpable.

Temporary abandonment: The well status where the well is abandoned or/and the well control equipment is removed with the intention that the operation will be resumed in the well within a specified time frame (from days up to several months).

Pore pressure (PP): Pore pressure is the pressure of the fluid inside the pore spaces of a formation.

Well integrity: A quality or condition of a well in being structurally sound with competent pressure seals by application of technical, operational and organizational solutions that reduce the risk of uncontrolled release of formation fluids throughout the well life cycle.

Waiting on cement (WOC): Waiting on cement, normally expressed in hours, is the period of time after the cement has been placed until the time subsequent drilling or completions operations can resume.

Well Security: Ensuring that the well does not go underbalanced or does not get into loss circulation at any point during well construction including cementation and WOC.

**CEMENTING OPERATIONS****4.0 CEMENTATION REQUIREMENTS****4.1 CASING CEMENTATION**

- i. The cement when set, shall be capable to provide lasting structural support and zonal isolation.
- ii. Cement slurries used for isolating permeable and abnormally pressured gas bearing zones shall be 'gas tight' for good zonal isolation and preventing annular gas migration problem.
- iii. Temperature exposure, cyclic or developing over time, should not lead to reduction in strength or isolation capability of the cement sheath.
- iv. The cement placement technique used should impose minimum overbalance on weak formations. Risk of lost returns and well activity during cementing shall be assessed and mitigated. Software simulation should be carried out for ensuring well security in deeper casings and all critical jobs.
- v. Centralisation plan and Cementing plan shall be issued for each cementation job to IM/ OIM and other associated persons (like chemist, geologist, mud logger, barge engineer, etc.) at field as applicable.
- vi. Cement height in casing annulus:
 - a. For surface casing, cement height shall be based on load conditions from wellhead equipment and operations. Top of cement (TOC) should be inside the conductor shoe or up to surface / seabed if no conductor is installed. Surface casings are cemented with cement top at seabed for subsea deepwater wells.
 - b. For casing through hydrocarbon bearing formations, cement height shall be based on requirements for zonal isolation. Cement should cover potential crossflow interval between different reservoir zones. Good cement barrier should be present 5 to 10 m above and below the point of potential inflow/ formation with hydrocarbons. If cement rise up to previous casing shoe is not planned, well has to be designed accordingly.
 - c. If the cement column required is over 1000m, two stage cementation should be planned with due risk assessment.
- vii. Cement verification:
 - a. The set cement sealing ability should be verified through a formation integrity test after a window of 2-3 feet is drilled out after the casing shoe is drilled out.
 - b. The cement height shall be
 - Verified by logs (cement bond, temperature, LWD sonic), or
 - Estimated on the basis of records from the cementing operation (volumes pumped, returns obtained during cementing, etc.).

**CEMENTING OPERATIONS**

- c. The strength development of the cement slurry should be verified by UCA/ HPHT curing chamber using cement slurry prepared with representative samples from the field and cured under representative temperature and pressure conditions in cement laboratory.
- d. Zonal isolation shall be verified by cement bond evaluation logs like CBL/VDL, USIT etc.

4.2 PLUG CEMENTATION

- i. A cementing plan shall be issued for each cement plug job to IM/ OIM and other associated persons (like chemist, geologist, mud logger, barge engineer, etc.) as applicable.
- ii. The properties of the set cement plug shall be capable to provide lasting zonal isolation.
- iii. Cement slurries used in plugs to isolate permeable and abnormally pressured hydrocarbon bearing zones shall be designed to prevent gas migration.
- iv. Permanent cement plugs shall be designed to provide a lasting seal under the expected static and dynamic conditions and loads downhole.
- v. Minimum batch volume of cement shall be defined for the plug to ensure that the homogenous slurry can be made, taking into account the contamination.
- vi. The length of firm plug shall be minimum 60 m. If a plug is set inside casing and with a mechanical plug as a foundation, the minimum length of plug should be 30 m. However, minimum 15 m cement plug should be left above the mechanical plug if further operation is planned above the abandoned section.
- vii. The plug shall extend minimum 30 m (measured depth) above any source of inflow/ leakage point. A cement plug in transition from open hole to casing should extend minimum of 30 m inside casing and at least 30 m (measured depth) below casing shoe.

4.3 PLUG VERIFICATION:

- i. The strength development of the cement slurry should be verified by UCA/ HPHT curing chamber using cement slurry prepared with representative samples from the field and cured under representative temperature and pressure conditions in cement laboratory.
- ii. The plug installation shall be verified through documentation of job performance; records from cementing operation (volumes pumped, returns obtained during cementing, etc.).
- iii. Each cement plug in the cased hole shall pass the following tests to verify plug integrity:
 - A pipe weight of at least 15,000 pounds/ 7 tons on the plug; or
 - A surface pressure of at least 1,000 pounds per square inch. For surface casing plugs, test pressure shall be 500 psi
 - The test pressure shall not exceed rated casing pressure, less casing wear factor.
- iv. Each Reservoir isolation plug in cased hole shall pass the following tests to verify plug integrity:

**CEMENTING OPERATIONS**

- A pipe weight of at least 15,000 pounds/ 7 tons on the plug; and
- A pump pressure of at least 1,000 pounds per square inch.
- v. The test pressure must not drop more than 10 percent in 15 minutes and then remains constant.
- vi. For plug in open hole, it shall be tagged / dressed to confirm depth of the firm plug.
- vii. If a mechanical plug is used as a foundation for the cement plug, the cement plug on top of it need not be verified if the mechanical plug is tagged and pressure tested.
- viii. For radioactive material left in the well, a cement plug of minimum firm length of 50 m and specific gravity as per guidelines of Atomic Energy Regulatory Board, Govt of India shall be placed over the location where the source is stuck.

5.0 OIL WELL CEMENT**5.1 OIL WELL CEMENT SELECTION CRITERIA**

Oil Well Cement should be as per API Specification – 10 A on “Specification for Cements and Materials for Well Cementing”/ ISO 10426-1:2009 on “Petroleum and Natural Gas Industries- Cements and materials for well cementing - Part1: Specification”/ IS 8229 (1986): Specification for Oil-well Cement or equivalent.

5.2 MARKING AND PACKING

Marking and packing should be as per API- Specification - 10A on “Specification for Cements and Materials for Well Cementing”/ ISO 10426-1:2009 on “Petroleum and Natural Gas Industries- Cements and materials for well cementing - Part1: Specification”/ IS 8229 (1986): Specification for Oil-well Cement or equivalent. The guideline should however be suitably modified to align with government guidelines on packing material issued from time to time.

5.3 HANDLING AND STORAGE OF CEMENT AND ADDITIVES**5.3.1 Handling**

- i. Care shall be taken during handling and transportation of cement/ additive bags so as not to allow ingress of moisture.
- ii. Use of iron hooks should be avoided as it will puncture the bags and expose the material to moisture.
- iii. Contamination of cement by other bulk materials should be avoided in offshore operations.
- iv. Air used for bulk transferring of cement / additives should be dry.
- v. For offshore use, cement additives should be sent to the installation in sealed containers. In case cement additives are sent on pallets, these should be properly covered to avoid ingress of moisture.

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- vi. Material Safety Data Sheets (MSDS) for cement and all additives shall be readily available, displayed and clearly communicated to all relevant personnel.

5.3.2 Storage

- i. Cement chemicals should be kept separate from other chemicals to avoid any intermixing of bags.
- ii. Storage of cement / chemicals should be such that oldest stock be preferably issued first.
- iii. Cement chemical bags should be stored away from the wall on a platform with approximately 6 inches air space and tarpaulin should be spread on the base of stacking in order to form a moisture proof seal.
- iv. While storing cement for a long time in silo, either on site or in bulk handling plants, cement should be transferred from one silo to another, at least once in a month.

6.0 CEMENTING EQUIPMENT**6.1 CEMENTING UNIT**

- i. The deployed cementing unit is designed to -mix and deliver as accurate as possible volume and density of cement slurry with the necessary properties to secure the casing and ensure integrity of zonal isolation. The pumps and associated lines should have rating to handle 1.5 times the maximum anticipated pumping & displacement pressures.
- ii. Pressure release safety valve shall be placed in discharge line so as to avoid sudden rise in pressure in the unit.
- iii. Recirculating system should be part of the cementing equipment or otherwise batch mixer should be used, particularly for cementation of the production casings.
- iv. Cement slurry density should be monitored by Densometer / pressurised mud cup balance before pumping into the well.
- v. In case of offshore rigs, the unit should have two centrifugal pumps for mixing water, one directly coupled with the main transmission and other with independent drive motor.
- vi. If the cementing unit with associated systems is intended to function as backup for the drilling fluid circulating system, it shall have required discharge and working pressure rating to meet the dual purposes.

6.2 CEMENT SILOS AND MOBILE PNEUMATIC BULK CARRIER

- i. The cement silos should be able to store sufficient bulk cement for meeting the maximum requirement of cement during the stages of well construction.

**CEMENTING OPERATIONS**

- ii. Each silo should be designed such that during cementation operations, discharge of dry cement and cement slurry to open environment should be minimized and meet the stipulated requirements.
- iii. Regular health check of silos and associated piping/ valves shall be ensured as per OEM.
- iv. The safety valves installed on the silos shall be calibrated in accordance with OISD-STD-106.
- v. There shall be suitable provision to know quantity of cement in silo.
- vi. All the inlet, outlet and vent lines from the silos should always be kept clean to avoid any blocking /choking.
- vii. In line filters should be installed on the inlet and outlet ports of silos to avoid any cement stones entering into the system. These filters should be periodically cleaned.
- viii. Compressed air used for transfer of cement from silos to surge tank should be free of moisture. Air release from the vent line should be passed through strainers / filters / dust collectors to arrest release of cement particles into atmosphere.
- ix. Regular health check of mobile pneumatic bulk carrier and associated piping/ valves shall be ensured as per OEM.

6.3 LOADING POD

- i. The loading pods shall be designed such that during cementation operations, discharge of dry cement and cement slurry to open environment should be minimized and meet the stipulated requirements.
- ii. Regular health check of loading pod and associated piping/ valves shall be ensured as per OEM.
- iii. Loading pod shall be equipped with a screw conveyor with a hopper placed at the ground for loading otherwise the tank should be provided with a suitable ladder.

6.4 CEMENT SURGE TANK

- i. The cement surge tanks shall be designed such that during cementation operations, discharge of dry cement and cement slurry to open environment should be minimized and meet the stipulated requirements.
- ii. Regular health check of surge tank and associated piping/ valves shall be ensured as per OEM.
- iii. The surge tank should be equipped for controlled discharge of cement to the mixing hopper.
- iv. It shall be equipped with safety valve and level indicator.
- v. Surge tank should be kept empty after each cement job. Provision should be made to back load the left-over cement from surge tank.
- vi. Vent line of surge tank should be unrestricted for smooth flow of cement from Silos to Surge tank.

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6.5 CEMENTING HEAD

- i. The cementing head should be cleaned, serviced, greased before and after every cement job.
- ii. Cementing head should be equipped with positive plug release indicator. Plug release indicator and assembly should be serviced regularly to ensure its proper functioning at the time of job.
- iii. Health check-up including pressure testing of cementing head shall be as per OEM recommendations.
- iv. HP valves on cementing manifold shall be tested periodically at rated working pressure.
- v. 'O' rings should be checked before every job for any cuts or damages.

6.6 ADDITIVE MIXING TANK / SLUG PIT

- i. For wet mixing of cement additives, a separate tank should be provided with suction facility from bottom and with delivery line to cementing unit displacement tank. Otherwise, the mixed water should be agitated by circulation.
- ii. The tank should have an agitator/ stirrer/ jetting gun for proper mixing and to avoid settling of additives in the tank.
- iii. The tank should be provided with proper facility for disposal of residual mixing water which will be disposed in accordance with OISD-RP-201.

6.7 HIGH-PRESSURE LINES AND MANIFOLDS

- i. All high-pressure lines and cementing manifold should be of pressure rating of minimum 1.5 times the maximum anticipated pressures during the cement job.
- ii. Pipeline layout should have minimum bends so as to minimise pressure loss.
- iii. For offshore rigs there should be two high-pressure lines from the cementing unit to derrick floor.

7.0 PRIMARY CEMENTING PRACTICES

Major areas requiring attention are:

- i. Slurry design / job planning
- ii. Blending of cement with additives in bulk handling plant
- iii. Casing lowering
- iv. Pre-cement job arrangements
- v. Job execution
 - Cement slurry mixing and pumping
 - Displacement

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- Waiting on cement
- vi. Post job analysis
- vii. Prevention of sustained casing head pressure

7.1 CEMENT SLURRY DESIGN / PLANNING OF CEMENT JOB

- i. The following information should be available for slurry design and cementation plan:
 - a. Rig, Well No., field location
 - b. Type of job - casing or liner: conventional or sub-sea operation.
 - c. Casing sizes, grades, weights and threads.
 - d. Casing depths, deviation, hole size, calliper log and drilled depth.
 - e. Type of mud used, mud parameters and rheology.
 - f. Bottom Hole Circulating Temperature (BHCT) & Bottom Hole Static Temperature (BHST) expected pore pressures and fracture pressures.
 - g. Interested zone intervals with Oil Water Contact (OWC) and / or Gas Oil Contact (GOC).
 - h. Well problems; anticipated / faced, during drilling (high pressure gas, Lost circulation etc).
 - i. Cement rise needed in the annulus.
 - j. Cement additives available.
 - k. Any other specific requirement like gas tight, resistance to cyclic temperature / pressure fluctuations, high cement strength etc.
- ii. At job planning stage, the availability of equipment and additives for the job execution should be assessed.
- iii. Calculate the cementing parameters which include quantity of cement, total mix water, pump rate, well head and bottom hole pressure during displacement, mixing time, mud displacement volume, surface pressure and other related information.
- iv. Design for maximum allowable down hole slurry density to prevent fracturing. Slurry density should be preferably 2-3ppg/ 0.24-0.36 sp. gr. heavier than the drilling mud.
- v. Determine BHCT from logs and using API Temperature data/ software assisted temperature simulation. Temperature logged approximately 24 hours after the last circulation can be used as BHST for API table use.
- vi. Fluid loss of cement slurry (as per test given in API Specification –10 A) should have values as given below:
 - a. For preventing gas channelling: less than 50 ml / 30 minutes.
 - b. For Liner Cementing : less than 100 ml / 30 minutes.
 - c. For casing cementing : approximately 250 ml / 30 minutes.

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- vii. The design should provide displacement in turbulent flow to obtain minimum 10 minutes' contact time at the top of pay zone. Preflush i.e. aqueous solution of dispersant and surfactants are not recommended in high pressure wells.
- viii. Weighted spacers / buffers should be used in high-pressure wells and preferably be designed for turbulent flow. The best results are obtained if not only the density but also the rheological properties of the spacer fall between those of mud and cement slurry. Design the spacer so that it is compatible with both the cement slurry and the drilling mud.
- ix. If turbulent flow is not possible, then design the job for mud displacement in effective laminar flow using rig pump rate as fast as possible within the limitations of fracture gradient.
- x. Minimum 35% Silica by weight of cement should be added to cement slurries in primary cementations for BHST above 230° F (110°C) to prevent strength retrogression of set cement.
- xi. Silica can be used as additive up to maximum BHST of 400° C. However, in case of wells subjected to temperature cycling as in thermal wells, oil well cement with Silica cannot be used above 320°C as it disintegrates. Calcium- Aluminate cement can be used in such conditions.
- xii. For normal slurries, control free water separation to 1.4% or less. To prevent gas channelling and in highly inclined wells, zero free water should be primary objective.
- xiii. Determine cement slurry thickening time at BHCT and bottom hole pressure. Minimum thickening time should be job time plus one-hour to attain a consistency of 50 Bc. Excessive slurry thickening time should be avoided.
- xiv. In wells, where field slurries are to be batch mixed, while running the thickening time test in lab, before following API test schedules for increasing temperature and pressures, the slurry should be stirred in the consistometer at the surface temperature and atmospheric pressure for the estimated batch mixing and holding time prior to pumping slurry into the well.
- xv. Slurry consistency for normal turbulent flow should be 10 to 20 Bc.
- xvi. Use the same mixing water, cement and additives in laboratory testing that will be used at the well for the job. If additives are dry blended, a blended cement sample should be collected and tested in the cement test lab to confirm slurry properties before the execution of the job.
- xvii. In case of certain types of cement jobs, such as long cement column rise or long liners where the static temperature at the top of cement is much lower than the BHCT, thickening time should be determined at the TOC temperature also.
- xviii. Minimum compressive strength of set cement at BHST should be:
 - a. For drill out: 500 psi
 - b. For perforations:
 - > 1000 to 1500 psi for non-HPHT wells

**CEMENTING OPERATIONS**

> 2000 psi or higher for HPHT wells and in wells where high pressure fluctuations are expected during hermetical testing, well activation, thermal stresses from steam injection etc.

- xix. Tensile requirement of cement sheath should be assessed and included as a parameter of slurry design in wells where the casing may be exposed to high stresses during life of the well wherever feasible.
- xx. Sufficient WOC time as per lab test report should be provided for light weight cement slurry placed around surface casing in deep sea wells due to slow hydration rate under low temperature environment.
- xxi. Cement additives such as fluid loss, dispersant, retarder, accelerator, gas block, expander, light weight and heavy weight should be added in order to get the desired properties in the cement slurry as per job requirement.
- xxii. For primary cementation of high-pressure gas wells, gas block additives should be added to cement slurry for effective blockage of gas leakage and the slurry should be tested for its gas tight property in the lab prior to use in the field.
- xxiii. During design of cement slurry, long term durability of the set cement subjected to stresses due to temperature and pressure cycling during the life of the well should be taken care of to prevent annular pressure build up problems.
- xxiv. Casing load is maximum when cement slurry is still inside the casing. While selecting casing, this aspect shall be considered.
- xxv. Large size surface casing with long column of cement in the annulus will have a tendency to float. Arrangement should be made for securing the casing.

7.2 BLENDING OF BULK CEMENT WITH ADDITIVES

- i. Cement absorbs moisture from its surroundings, which adversely affects its properties. Therefore, cement should be purchased with staggered deliveries
- ii. During blending of cement with dry additives, verify weight calculations, quantity and name of each additive going into the cement with the design composition. Count the additive sacks and cement sacks for each blend.
- iii. Visual inspection of empty tanks is suggested prior to transferring blended materials to ensure that they are completely empty.
- iv. To ensure proper blending, transfer the blended material at least twice between silos before loading on mobile silos or bulk supply boats for drill site delivery.
- v. Conduct fluid loss and thickening time test on samples taken from each container to verify blending.

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- vi. Transfer dry blend to empty tank and back to original tank just prior to slurry mixing. If extra empty tank is not available on site, then fluff or percolate air through each tank from the bottom for 15-20 minutes to redistribute additives.
- vii. When different blends are used, each blend and corresponding tank should be clearly identified.
- viii. Collect and save sample of each composition for post job analysis, if failure occurs.
- ix. Check that all materials required for the job are loaded and identified on the transporting vehicles. Verify that correct chemicals are loaded with margin in weight / volume.

7.3 LOWERING OF CASING

- i. Running-in speed of casing should be controlled to prevent fracturing and mud loss. Casing lowering should be so regulated that the maximum annular velocity of drilling fluid caused by the movement of the pipe does not exceed the annular velocity during normal circulation. During lowering long casings and also in HPHT wells, intermediate circulation at previous casing shoe and gel break in open hole should be carried out to reduce incidences of mud loss during circulation on reaching bottom.
- ii. Casing should be filled with drilling fluid at regular interval as per lowering plan.
- iii. Last two casing joints should be lowered at a very slow speed. (and with circulation in rigs having Top drive system).
- iv. Landing joint(s) should be spaced out so that the cementing head can be installed easily at the rig floor after casing is landed.
- v. Casing centralisation is a critical parameter that should be part of the cementing programme. Placement of centralizers should be done with the help of cementing software in order to have casing standoff ratio near 75% for deviated wells and near to 100% for vertical wells. Welding of stop ring on casing and use of welded steel bows should be avoided. Centralizers with mechanical properties as per API/ equivalent standard should be used.
- vi. Scratcher may be used against permeable formations and 60 m above and below the pay and water zones in case of casing reciprocation.
- vii. Double float protection (a float shoe and a float collar) should be used. Check its functioning on surface before making up.
- viii. The shoe track should have enough volume to contain all cement contaminated by mud displaced from the casing by the top plug. As a minimum, assuming range 3 casing (average length of 38.5 feet), this should be two joints for large ($\geq 13\text{-}3/8$ inch) diameter casing and at least three joints for smaller inside diameter (ID). Longer shoe track length is planned as set depth of the casing strings increases.

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- ix. In case of differential float equipment, check the size of the ball for actuating them properly.

7.4 PRE-CEMENT JOB ARRANGEMENTS**7.4.1 General considerations**

- i. A detailed plan indicating the step-by-step job sequence should be prepared and duly acknowledged by the installation manager and circulated to all concerned prior to the cementation job.
- ii. A work permit, in compliance with OISD-STD-105, shall be obtained prior to commencing any cementing operations.
- iii. After arrival at site, the cementing in charge should independently calculate the slurry volume and displacement volume required using actual parameters and also total water requirement including the water needed to flush cementing unit and high-pressure lines.
- iv. Ensure that the supply rates of water required for the job and mud for displacement are sufficient for uninterrupted operation.
- v. When displacement is to be performed by the rig pump, availability of at least two rig pumps should be ensured. Also, the discharge rate per stroke, volumetric efficiency of the pumps, functioning of stroke counter and mud line flow meter should be checked.
- vi. Check air compressor operation, fluff the cement silos, prime the pumps & mixers and check for their operation.
- vii. Proper working of air dryer should be ensured, especially in offshore.
- viii. In case of floater rigs, a mandrel is used for carrying top and bottom plugs. Check its ball for correct size before installing. The mandrel should be serviced immediately after every use.
- ix. Check cementing head, plug release mechanism, plug dropping indicator and review the number and placement of rubber plugs in proper sequence.
- x. Inspect top and bottom cement rubber plugs before loading. Turn bottom plug upside down and inspect hollow core and rubber diaphragm. Do not puncture diaphragm of bottom plug prior to loading. Bottom hollow plug is loaded first, and then top solid plug is loaded. Ensure correct order of plug loading and locking of plug release shaft.
- xi. A bottom plug is not recommended with large amounts of lost circulation material in the well or with rusted or scaled casing, as such material may collect on the ruptured diaphragm and block the flow through the casing.

7.4.2 Cementing equipment

- i. Hook up the cementing units, silos, surge tank, compressor and bunkers and check each component to avoid any breakdown during the operations.
- ii. Check the discharge of each cementing unit physically for their capacity in accordance with the liner size.

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- iii. Pressure test of pumping lines and valves should be carried out at the maximum pressure expected during the job with safety margin of 10-20%.

7.4.3 Mud conditioning

- i. Begin pipe movement and mud conditioning immediately after the casing is near bottom. The casing movement should continue throughout the circulation period. When reciprocating, the pipe is usually moved through a 4 to 6 m stroke, using a 2-minute interval for the cycle. For rotation, the pipe is rotated at around 10 rpm as per well condition. This is however not feasible in sub-sea wells.
- ii. Condition the hole with good surface conditioned mud at maximum possible rate within the limitation of formation fracture gradient for 1.5 to 2 cycles (minimum). Adjust Plastic Viscosity (Pv) and Yield Point (Yp) of the mud to the lowest possible values after one cycle.
- iii. For critical jobs, such as production casing cementation, freshly prepared treated mud having low rheology should be pumped ahead of spacer and cement slurry.

7.5 JOB EXECUTION

Conduct PJSM (pre-job safety meeting) with all operational personnel explaining their respective roles and responsibilities, critical operational safety aspects and emergency response.

7.5.1 Cement Slurry Mixing and Pumping

- i. Pressurise bulk units to 15-25 psi just prior to starting of slurry mixing.
- ii. Start the pumping operation with pre flush to establish circulation to ensure that the casing shoe is open and check the mud return.
- iii. Do not premix the cement additives in the water more than 5-6 hours before the cementing job. It is preferable to wait until the final circulation is started after casing lowering to the target depth. Verify metering device if liquid additives are premixed in water. Continue to agitate the chemical mixed water thoroughly until the job is complete.
- iv. The chemical water should be measured through displacement tank because it helps to calculate easily how much quantity of cement has been pumped in the event of unplanned shutdown.
- v. Control slurry density with online densometer / pressurised mud cup balance. Check calibration of densometer as well as mud cup balance with fresh water to ensure the reliability of density readings.
- vi. Use preflush or spacer equal to volume of 150-200 m annular height in the annulus being cemented. Pump Preflush or spacer ahead of bottom plug. Better prevention of contamination can be achieved by use of two bottom plugs, one ahead of Preflush and one ahead of cement slurry.
- vii. To ensure good control of slurry density and other properties, batch mix cement slurries. Alternatively use continuous mixing devices like Precision Slurry Mixer (PSM) or Recirculating Cement Mixer (RCM).
- viii. Do not open cementing head to drop top plug as it allows the well to suck in air and cause honey combing of cement around the shoe joints.

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- ix. Do not try to get the last quantity of cement out of the cement bunker or surge tank. This will cause reduction in slurry density and will result in poor slurry at shoe joint and in annulus of bottom casings.
- x. If the slurry density cannot be maintained within acceptable limit or there is interruption due to equipment break down making the operation risky, it is advisable to call off the job and circulate the cement slurry out of the well.

7.5.2 Displacement

- i. Determine displacement rate on the basis of the type of casing string to be cemented. Turbulent flow displacement is usually accepted as being the most efficient technique for achieving good mud removal. However, when turbulence all around the casing cannot be achieved mainly due to stand off, pump or fracture gradient limitations, then the maximum permissible / attainable discharge for high laminar flow is recommended during displacement.
- ii. Cement displacement rate should be as high as possible to attain annular velocity of at least 1.3 m/sec and preferably 1.8 m/sec to achieve good zonal isolation with reasonable degree of certainty.
- iii. Field validated computer programs should be used to calculate the highest possible displacement rate within the constraints imposed by formation strength and surface / downhole equipment limitation.
- iv. Casing pipe should be kept in motion to improve mud displacement. Reciprocation should be on a two-minute cycle over 4 to 6 m length. Rotation of around 10 rpm is satisfactory (if permissible).
- v. Release top plug from cementing head with minimum down time.
- vi. Continue pipe movement till around 95% of the displacement volume.
- vii. Calculate the anticipated mud return volume throughout the job.
- viii. Mud return is measured in trip tanks or by other means. High return rate is predicted during the free fall period due to U-tube effect.
- ix. Monitor the displacement and bumping of plug. Check the flow back by releasing the casing pressure. Leave casing open during waiting on cement (WOC). A small amount of back flow is expected because of thermal expansion.
- x. Back pressure applied in the annulus helps in containing formation influx during cement setting.
- xi. Even if the rig pump performs the displacement, then also cementing unit shall remain completely lined up until plug bumps. The plug hitting should be observed carefully and pressure reported in cementation job report.
- xii. In the event of failure of float valves, the casing should be kept closed at pressure equal to the differential pressure till the cement slurry thickens.
- xiii. During cementation, 'mud returns' from the annulus should be continuously monitored.
- xiv. Pressure tests the casing for hermeticity immediately after the top plug bumps in cases where the displacement fluid is water, which is feasible in low mud weight wells and this practice may be adopted in sub hydrostatic/hydrostatic oil wells to reduce time prior to perforation.

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- xv. Maintain log of operations to include operation in progress, time, density measurement, mixing rate, volume of fluid pumped, pumping pressure, displacement rate etc.
- xvi. All specific events occurring throughout the cement job should be recorded along with other relevant points for post job evaluation.
- xvii. Carry out material balance calculations for mix water, cement and cement additives and compare with volume of each slurry pumped.
- xviii. Prepare a summary of the completed job.

7.5.3 Waiting-On-Cement (W.O.C.)

- i. Based on slurry design and application, W.O.C time shall be mentioned in the cementing plan which is required -for the cement to develop adequate strength before operations are resumed. During the W.O.C time, the cement though rigid, has very little strength, and any damage sustained by the cement sheath during this period does not 're-heal'.
- ii. The required period of W.O.C time varies depending on the cement blend and downhole condition of temperature and pressure.
- iii. Cement used in the casing annulus should have compressive strength of at least 500 psi before drilling is resumed.
- iv. Cement should have compressive strength of 1000 to 2000 psi before the well is perforated, as described in clause 7.1 (xviii).

7.6 POST JOB ANALYSIS

Evaluation of cement job is very crucial to determine its success. A complete post job analysis comparing field job parameters with evaluation results is the best way to reasonably understand what happened downhole and accordingly necessary corrective measures for future operation should be applied.

- i. CBL-VDL, CET or USIT or equivalent logs can provide valuable information regarding the cementation job.
- ii. The cement bond log, normally, should not be run until 48 hours after the cementation in order to achieve the true cement bond reading. This again is highly dependent on the cement type and additives used in the slurry and bottom hole conditions.
- iii. In case cement bond is found inadequate, it is suggested to re-run cement bond log (700/1000psi) after 24 hrs of first log prior to decision on cement bond repair by cement squeeze, which is time consuming and also compromises the casing integrity.
- iv. Field results show that most of the wells exhibit a micro-annulus on a primary cement job. Cement evaluation logs should be recorded under 700-1000 psi pressure to eliminate micro-annulus effect, wherever feasible.
- v. The bond index method is used for interpretation of amplitude curve in cement evaluation logs towards achievement of zonal isolation. This method is essentially a graphical solution, which allows determination of amplitude value corresponding to a particular bond index.

The guidelines as per best industrial practices based on the bond index method is given below:

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Satisfactory 'length & bonding' required for zonal isolation for different size casings:

5" Casing	-	5 feet of 60% bond
5-1/2" Casing	-	6 feet of 60% bond
7" Casing	-	10 feet of 60% bond
7-5/8" Casing	-	12 feet of 60% bond

7.7 PREVENTION OF SUSTAINED CASING HEAD PRESSURE

The main objective of cement sheath in oil & gas well is to prevent any fluid communication during the life of the well so that the reservoir can be produced safely and economically. To achieve this objective, during cementation, the drilling fluid should be displaced completely by the cement slurry in the annulus till TOC. However sustained casing pressure observed on a number of wells in the old fields around the globe has led to the conclusion that even if the slurry was properly placed during the cementing job, and initially fulfils its isolation role, changes in downhole conditions can induce sufficient stresses to destroy the integrity of the cement sheath. The consequence will be loss of zonal isolation, leading to long-term gas migration problem. Therefore, besides placement, the set cement should be able to withstand the stresses induced by the well events and maintain its integrity during the life of the well.

Studies indicate that in the long term, more important than compressive strength is the ductility / flexibility of the set cement to withstand downhole stresses imposed by pressure and temperature oscillations during drilling, production and workover operations. Cement systems that are stiffer or possess a high Young's modulus are more susceptible to damage when subjected to changes in pressure and temperature. Therefore, for long term durability, the set cement should have high tensile strength to Young's modulus ratio, low Young's modulus value compared to that of rock and higher Poisson's ratio. Also in a weak formation, chances of cement sheath failures are more, as it cannot mechanically support the cement deformation.

Examples where well integrity can be lost due to cement failure because of high stresses are

- Deep water wells
- HPHT wells
- Wells completed in weak unconsolidated formations
- Steam injection / cyclic steam stimulation (CSS)/ water & gas injection/ in-situ combustion wells

The following are the stresses induced by different type of events during the life of a well:

- a. Large increase in wellbore pressure
 - Pressure integrity test
 - Increase of mud weight
 - Casing perforation
 - Stimulation
 - Gas production
- b. Large increase in wellbore temperature
 - HPHT wells
 - Deepwater wells

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- Thermal recovery wells
 - Geothermal production
- c. Formation loading
- Creep
 - Faulting
 - Compaction

Traditionally, cement sheath design in the majority of wells was focused on short-term properties and concentrated only on compressive strength as a quality indicator. The effects of well events and their subsequent effect on the cement sheath during the well's life have not normally been considered. In general, no attention has been paid to mechanical properties of set cement such as tensile strength, Young's modulus etc. This approach could serve well if the sealant is not subjected to a large change in stress level. But in actual conditions wells are exposed to many changing conditions that create mechanical stresses on the casing and the cement sheath behind it. These stresses can come from pressure changes; fluid weight changes during drilling and completion; pressure testing and pressure treating such as squeeze cementing or high-pressure stimulation treatments; changes in well pressures caused by reservoir pressure depletion. Temperature changes, especially in upper portions of a well producing high-temperature fluids, can also generate mechanical stresses.

Design procedures should be developed to estimate the risk of cement sheath damage over time as a function not only of the cement sheath properties, but also of those of the formation, casing and representative well events. Designing cement to maintain seal in the well bore throughout the life of the well should be based on information regarding the rock and casing properties, well geometry and future loading which are expected during the life of the well, as listed below:

- a. Physical and thermal properties of cement, casing and formation
- Young's modulus
 - Poisson's ratio
 - Tensile strength
 - Thermal expansion
 - Thermal conductivity
- b. Formation in-situ stresses
- c. Well design
- d. Representative well events

Software should be used to predict the stresses to be experienced by the cement sheath throughout the life of the well due to the changing conditions of the well. This software also assesses the mechanical properties of the cement to determine if the cement will survive these stresses. Thus, the mechanical properties of the cement can be modified so that the cement will survive to provide isolation for the life of the well. This methodology allows custom fit cement designs for individual well bore stress environments.

**CEMENTING OPERATIONS****8.0 SPECIALISED CEMENTING OPERATIONS****8.1 MULTISTAGE CEMENTING**

Multistage cementing comprises of conventional placement of cement slurry around the lower portion of the casing string followed by placement of cement slurry in successive upper stages through ports of stage collar. Mostly it is in two stages, although additional stages are possible. This technique is useful if long cement column is required or in cases of low ECD margin.

- i. Design cement slurries for the first stage and second stage at temperature and pressure at the casing depth and stage collar depth respectively.
- ii. Before the stage cementing collar is made up in the casing string, check the size of the trip plug/freefall plug/opening bomb, internal diameter of the opening sleeve and closing sleeve and the size of the seats provided in opening & closing sleeves.
- iii. Regardless of the type used, caution must be exercised in the initial handling of the stage collars, as the equipment is manufactured to close tolerances. Smooth sliding and sealing of the concentric sleeves is necessary for proper operation. Rough handling prior to or during installation can “egg” or misalign the moving parts, causing failure during job execution.
- iv. Compatibility of float collar and the stage collar should be ensured. The first-stage wiper plug (if used) and the first-stage displacement plug must fit and seal against the float collar.
- v. Run in the casing with the stage collar at the desired depth. Stage collar should be tightened by putting the tong at the designated place on the outer sleeve of the stage collar only. The stage collar should preferably be placed against the shale section.
- vi. Besides slurry displacement time and safety factor, the first stage slurry thickening time should include the travelling time of bomb, opening of ports and one cycle circulation through stage collar ports.
- vii. If there is requirement of positioning stage collar in high inclination section of the well, hydraulic stage collar should be used. Circulating pressures during first stage should be maintained well below recommended sleeve opening pressure.
- viii. At least one centraliser each should be placed just above and below the stage collar.
- ix. After completion of first stage cementation job, drop the opening plug or bomb and allow it to reach the stage collar. Opening plug velocity is approximately 1 m/sec in normal mud. Ensure the seating of free fall plug for opening of stage collar port. To open the ports, slowly build-up pressure usually approximately 1200 to 1500 psi. A drop in pressure indicates that fluid has entered - into the annulus after opening of ports.
- x. Circulate the well through the stage collar ports for 2 cycles to flush out any extra / contaminated cement from the first stage and the well shall be circulated until the mud is conditioned for the second stage.
- xi. In case of loss prone sections, pumping of cement slurry for second stage should be resumed after thickening time of first stage cement slurry to avoid loss/ activity.

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- xii. Release closing / shut off plug in such a manner so as to have some cement slurry over it. This will ensure uncontaminated cement behind- the stage collar. Further, cement above shut off plug will ease drilling of the plug.
- xiii. For closing the cementing ports, recommended closing pressure in excess of second stage final pressure should be built up over the shut off plug in one continuous operation without slowing down or stopping the pumps. Confirm the closure by bleeding back.

8.2 STAB IN / INNER STRING CEMENTING

Stab in/ Inner string cementation should be considered in deeper set large size (≥ 20 inch) casings due to following advantages:

- i. By pumping through the smaller diameter inner string,
- ii. Less time is required for circulation
- iii. Cement contamination resulting from channelling inside the casing is reduced
- iv. Better control over displacement
- v. Shorter displacement volume / time. So shorter cementation time.
- vi. Large diameter cementing head and rubber plugs are not required
- vii. Cement is discharged to casing annulus much faster after mixing, reducing the risk of the cement slurry setting within the casing
- viii. Reduces the amount of cement that has to be drilled out of large-diameter casing.
- ix. In cases where TOC is planned to surface, slurry pumping can be stopped when good slurry appears on surface (and displace the slurry in drill pipe) rather than pumping excess slurry as per thumb rule.
- x. Shorter thickening time of slurry and shorter WOC.

The potential impact of annulus bridging on casing collapse should be considered in inner string cementing. The severity of this risk increases with longer/deeper casing due to longer preparation time.

8.3 LINER CEMENTATION

A liner is a standard casing string that does not extend all the way to the surface up to the well mouth, but it is hung from inside the previous casing, generally keeping an overlap of 50 to 150 m.

8.3.1 Liner Hanger with metal-to-metal sealing may be considered for HPHT wells or long liners. Liner which is rotated & reciprocated till end of major displacement will result better cement in annulus. Hanger sealing is robust with metal-to-metal sealing. Hanger top clean out (cement drilling) can be eliminated as TOC is not required up to hanger top.

Selection of liner hanger should be done based on risk assessment and cost benefit analysis.

- i. Slurry Design: While designing the cement slurry for liner cementation job, the following parameters should be considered:
 - a. Thickening time: Time taken for washing out the excess slurry above the liner hanger top should be considered while designing slurry. However, the wells where high pressure gas is

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being isolated behind the liner, relatively short thickening and setting time are necessarily required to reduce chances of gas penetrating the unset cement.

- b. Slurry Density: High-density low water ratio slurry should be used to prevent water separation and entry of fluid into the wellbore, but the combined density and displacement pressure must remain below the fracture pressure of the weakest zone.
- c. Fluid Loss Control: Fluid loss of the slurry should remain less than 100 ml / 30 minutes so as to avoid building up of cement filter cake and to reduce chances of annulus bridging due to higher pumping pressures resulting from small annular channels.
- ii. A four-arm calliper should be run prior to the liner operation to ascertain the hole size for calculation of cement slurry volume which is very critical for liner cementation.
- iii. No LCM (except system LCM) should be used in liner cementation to avoid plugging of float equipment or the narrow annulus. If LCM is added in mud system to combat losses, the well should be circulated with fresh mud free from LCM prior to cementation.
- iv. Centralising the liner in the hole is very critical, because annular clearance is so small that the liner must be kept clear of borehole wall for effective cement placement. This is particularly true in case of deviated wells. Bow spring centralizers should be used in the open hole if there is sufficient annular clearance. Rigid centralizers should be used in the casing/liner lap region, and also in the open hole in cases of small annular clearance. Centralizers or positive standoff devices also reduce the likely hood of differential pressure sticking of the liner in the open hole.
- v. In deviated wells rotating type liner is recommended to facilitate mud removal and placement of cement slurry at the lower side of the hole.
- vi. Use combination dart in case combination string is used for lowering liner to avoid bypassing of cement slurry.
- vii. In case of loss prone areas, preferably use liner hanger with integral packer or with top seal to avoid hanger top squeeze job.
- viii. The small clearance also makes it difficult to run liners. Swab/ surge pressure can be extremely severe and running speed should be such as to avoid pressure surges that could break down formations leading to lost circulation. -
- ix. Circulation should be carried out before setting the liner to clean the mud system of any cuttings or debris. Cuttings, if not removed can accumulate at the restricted area of liner hanger leading to bridging of annulus & rise in pressure during circulation & cementation. The circulating pressure should be monitored and restricted up to shear pin rating of hanger till hanger is set.
- x. Where bottom plug is not used in liner cementation, a spacer fluid compatible with mud and cement should be pumped between mud and cement to provide a buffer to avoid contamination.
- xi. The amount of excess cement for liner cementing must be carefully calculated by taking into account the well conditions and mud removal efficiency. Mud removal efficiency is a key variable in determining cement slurry volumes as it is not uncommon to have 60% to 80% mud removal in liner cementing.

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- xii. The volume of cement slurry used in most deep liners is usually rather small. Since slurry design is critical for liner cementation, strict quality control of the slurry shall be ensured during cementation.
- xiii. Displacement rate should be slowed down when the pump down plug (dart) approaches the liner wiper plug in order to observe the first pressure surge (about 300psi) corresponding to the shearing of the pins. Displacement volume till pressure surge point indicates the pump efficiency, which can be a guide for the remaining displacement.
- xiv. Pull the setting tool free from the liner and wash out any excess cement above the liner top. If no packer is incorporated into the liner hanger, then wash out keeping excess cement over the top of the liner so that 5 to 8 joints of the intermediate casing will contain cement to be drilled out after setting.
- xv. Reverse circulation places an extra pressure on the annulus and this additional pressure should be pre-calculated and controlled to avoid formation break down. A liner packer, once set, keeps reverse circulation pressure off the formation.
- xvi. In long liners, there may be a considerable temperature differential between the bottom and top of the liner. The cement may take very long time to set at the top and as such drilling of cement should be done after the cement sets- at the top end of liner also.

8.3.2 Testing of liner top

A leaking liner top can become a serious and expensive problem during future operations like production life of the well. Therefore, top of the liner, after it has been cemented should be tested for successful completion.

Two methods can be used to test the pressure integrity of a cemented liner top.

a. Hydrostatic testing

Testing the liner top with applied pressure can be done with or without a packer. However, in either case, burst limitation of the intermediate casing must be considered. In case of a drilling liner, pressure applied to the liner top should be equal to or greater than the hydrostatic pressure at the liner top when the maximum anticipated mud weight will be used in subsequent drilling operations. To ensure the pressure integrity of liner top, fracture gradient of the formation at the shoe of the intermediate casing must also be considered. Also, the test pressure should not exceed the hermetical test pressure of the casing.

b. Differential testing

For this test, a negative pressure equal to differential pressure that the well may encounter later in drilling or completion should be applied. For high-pressure gas wells, differential pressure test should be carried out in stages.

9.0 CASING INTEGRITY TEST / HERMETICAL TEST:

Integrity of the casing is tested at different phases of drilling and completion to ensure that the casing has no leakage and there is no possibility of influx from formation or well fluid loss to annulus from the casing

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in the given set up. This is also termed as hermetical test of the casing and carried out after primary casing cementation, cement plug and squeeze jobs. Casing integrity test or hermetical test is carried out by applying surface pressure (positive test) and after displacing the required length of well fluid with water or any clear fluid or gas of desired specific gravity thereby reducing the hydrostatic head in the well (also called inflow test or negative test). Some aspects of inflow test are listed here.

9.1 DISPLACEMENT OF WELL VOLUME TO LIGHTER FLUID DURING INFLOW TEST / COMPLETION

Inflow tests are used to test a barrier in the direction of flow. Since the barrier is not bidirectional in most cases, an inflow test is needed when the barrier cannot be adequately tested by applying positive pressure from below.

Barriers that will need an inflow test prior to well activation:

- Liner hanger/liner top packers
- Production Casing- to establish casing integrity
- Casing shoe track.
- Mechanical barriers inside casing
- Xmas tree valves, Downhole safety valves
- Casing hanger seals
- Packers inside casing

A well should be displaced to a light fluid (less than kill weight) during inflow test / completion operations. The active barrier is no longer the fluid column but the casing or liner and any mechanical barrier in the well.

The displacement should be planned with a step-down chart of pressure versus volume. During displacement, the volume of fluid returned should be monitored and compared with the volume of fluid pumped. Ideally there should be independent checks (e.g. mud loggers and rig crew).

If the barrier being tested fails when the well is in an underbalanced condition, there may be an influx or kick. Horner plots should be used to differentiate between pressure build up due to a leak and thermal effects.

There should be a contingency plan if there are indications of an influx during or after displacement operations. The well should be shut-in with secondary barrier in the well and the well circulated back to kill weight fluid under controlled conditions (through the choke system).

An inadequate inflow test or a 'false positive' (that is a test that is wrongly thought to have qualified a barrier) will lead to a major well control problem if the well is then displaced to light fluid.

Keeping the volume of lighter fluid to a minimum by pumping the fluid down the drill pipe run with a packer, rather than in the annulus, is preferable to achieve the above test objective for deep-set barriers.

Further it is reiterated that both internal pressures and drawdown could jeopardise cement sheath in the annulus of the well. So, caution should be exercised to limit the pressure/drawdown applied to verify the casing integrity.

In HPHT wells hermetical test of the production casing (deep wells with high mud weight) should be carried out with clear brine of reduced specific gravity to generate adequate drawdown. Displacement of

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the drilling fluid with water should be avoided. If displacement with water is unavoidable, partial changeover can be done in stages after setting RTTS packer at planned depths as per drawdown requirement.

9.2 SLURRY DISPLACEMENT WITH LIGHT FLUID

In case it is planned to displace the cement slurry with lighter clear fluid during cementing of a production casing and complete the hermetical test before cement in annulus sets, extreme caution needs to be exercised due to limitation of well control mechanism available during cementation. Since there is no inner string in the casing, well control will become very tricky in case of any influx. So, such plan may be adopted only in wells targeting sub-hydrostatic or hydrostatic oil-bearing reservoirs.

After displacing cement slurry with lighter clear fluid or water, positive test for casing integrity, if planned, should be carried out immediately (when cement slurry in annulus is still in fluid condition).

In all other wells, hermetical test should be carried out only after recording cement bond log.

9.3 SHOE TRACK AS A BARRIER

An internal (or positive) pressure test of the casing does not qualify the shoe track as a barrier, because typically the top wiper plug holds pressure from above but not from below.

Even if the shoe or collar float valves are pressure-tested before installation, they may be damaged or washed out during the cementing process.

The following actions will help to ascertain the adequacy of the shoe track as a barrier:

- Set a cement plug on top of the float collar and pressure test from above or
- An inflow test will provide a robust demonstration that the shoe track is an adequate barrier.

For further details, Refer to OISD-RP-238: Well Integrity.

For further details on drilling, cementing and testing of HPHT wells, refer to OISD-RP-242: Recommended Practices for drilling and testing of HPHT wells.

10.0 SECONDARY CEMENTATION PRACTICES

Secondary cementation jobs are mainly classified as

- i. Plug Cementing
- ii. Squeeze cementing

10.1 PLUG CEMENTING

Cement slurry of specified volume when placed and set across a selected interval in an open hole or a cased hole is called 'Cement Plug'. The most commonly used technique for plug placement is known as 'Balanced Plug Method'. Following the standard best practices as detailed below can prevent plug failures:

- i. Select a gauge section of the hole. Consult calliper log for selecting a location to set the plug and determine the temperature of the formation where the plug is to be set.
- ii. Circulate enough to condition the well so as to ensure that the entire mud is uniform.

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- iii. Check the mud system carefully for loss of returns, fluid gain or gas entry. Any movement of the plug after it is placed may cause the cement to remain unset.
- iv. A cement plug is best set against a competent hard rock. Shale should be avoided as they are often caved and out of gauge.
- v. However, if 'kicking off' is the objective, the plug should not be set in an excessively hard formation. Ideally, the plug should extend from a soft shale down to a hard formation. Logs and drilling rate records should be consulted when selecting a depth to set a plug for kick-off.
- vi. Slurry design
 - Viscous slurries with high gel strength and low density are needed for lost circulation plugs, to restrict flow into voids or fractures.
 - High compressive strength is required in kick-off plug to have a sharp contrast between the plug and the formation hardness. Use densified cement slurry that will tolerate considerable mud contamination.
- vii. Calculate cement, water and displacement volumes and always plan to use more than enough cement (1.5 to 2 times the calculated volume) to compensate for contamination effect so as to get the desired plug length.
- viii. Batch mix the cement slurry to ensure uniform slurry density.
- ix. Pump Preflush that is compatible with drilling fluid. Preflush volume should be sufficient to cover an annular height of 150 to 250 m and the after flush volume should cover the same height in the string as that of the Preflush.
- x. For open hole cement plugs in gas wells, use a weighted spacer 1-2 ppg (0.12-0.24 sp. gr.) heavier than the mud. Using water as Preflush can cause reduction of hydrostatic head resulting in gas influx and also migration through the cement.
- xi. Whenever possible Preflush / spacer should be pumped in turbulent flow conditions
- xii. Try to rotate or reciprocate the string slowly till the completion of displacement.
- xiii. Under displace the plug by 200-300 litres to avoid any back flow.
- xiv. Pull out the drill pipe/tubing slowly (10-15 m/minute) out of the cement to minimize contamination.
- xv. Reverse wash twice the string volume to wash excess slurry out of the hole. However, in case of narrow ECD margin, direct wash can be planned due to lesser pressure in the annulus during circulation.
- xvi. Ample WOC time should be allowed for a plug job. A common practice is to allow for longer WOC time since accurate well temperature for a cement plug job is difficult to determine.
- xvii. Always test the cement plug by tagging top of cement by applying weight of 15000 pounds/ 7 T.
- xviii. While placing a cement plug for kick-off special measures as given below should be followed for success at first attempt:
 - a. Use either a mechanical or chemical method to provide some static barrier below the intended bottom of the plug.

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- b. When a high viscous pill is used for achieving a static barrier below the cement column, then the length of the pill should be equal to the cement plug length and funnel viscosity of the pill should exceed 150 seconds. Also, the pill density should be greater than the mud weight and 10 seconds gel strength of the pill should be above 50 lbs/100 sq. ft.
- c. Use a 'Diverter tool' for placement of cement to achieve uniform placement of cement slurry all around the well bore and to prevent contamination due to downward movement.
- d. In case cement plug is placed in small dia. hole of 8-1/2" hole or lesser, a tail pipe should be connected in the string to minimize contamination of the cement plug when the pipe is being pulled out. The length of tail pipe should be 1.5 - 2 times the plug length. Pull out should be slow.
- e. For kick-off plugs, slurry should be properly designed and sufficient WOC should be provided for attaining required hardness/ strength.

10.2 SQUEEZE CEMENTING

Squeeze cementing is the process of forcing cement slurry, under pressure, through holes or splits in the casing / well-bore annular space and then allowing it to dehydrate by further application of pressure. Squeeze cementing is necessary for many applications, but the most important use is to segregate hydrocarbon-producing zones from the formations producing other fluids. A key element of a squeeze cementing job is placement of cement at the desired point(s).

In squeeze job, regardless of the technique used, the cement slurry (a suspension of solids) is subjected to differential pressure against a permeable rock that acts as a filter. The resulting physical phenomena are filtration, filter-cake deposition and, in some cases, fracturing of the formation. The slurry, subjected to a differential pressure, loses part of its water to the porous medium, and a cake of partially dehydrated cement is formed.

As the filter cake builds, the pump in pressure increases until a squeeze pressure less than fracturing pressure is attained. A good guide for a squeeze pressure is 500-1000 psi above the pump in pressure with no flow back in 3 to 5 minutes.

While carrying out squeeze jobs at shallow depth, application of pressure may result in upward force greater than the weight of string. Pressure limitation/ securing of drill pipe / tubing string should be considered before taking up such jobs.

In case of cement squeeze job through cement retainer, the integrity of string and seal should be ascertained by pressure testing it in hermetic position. If anticipated maximum squeeze pressure exceeds collapse pressure of the casing, pressure should be applied through string annulus to neutralise pressure acting on casing from outside.

10.2.1 Injectivity test prior to cement squeeze

Prior to placement of cement slurry, conduct injectivity test against the interval to be squeezed to determine if and at what rate below the fracture gradient, fluid can be placed against the formation. When

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the fracture gradient is to be exceeded to obtain sufficient injectivity rate for cement placement, it should be done with caution, without giving excessive pressure above fracture gradient.

A minimum of ten barrels volumes should be used when obtaining an injection rate. Deep perforations require more volume than shallow ones because of the additional hole volumes. Consider spotting a clear fluid such as water across the perforations when obtaining an injection rate. The injection test is performed for several reasons:

- To ensure that the perforations are open and ready to accept fluids.
- To obtain an estimate of the proper cement slurry injection rate.
- To estimate the pressure at which the squeeze job will be performed, and
- To estimate the amount of slurry to be used.

If suitable injection rate cannot be established at the desired injection pressure, it may be necessary to clean up the perforations, channels etc. Acids are commonly used for the purpose. While taking injectivity test, raise the pressure very slowly up to the point of injection without fracturing the exposed formation.

10.2.2 Design of cement slurry for squeeze job

The properties of cement slurry should be tailored according to the characteristics of the formation to be squeezed, and the technique to be used. Squeeze slurry should be designed to have the following general attributes:

- Low viscosity: to allow the slurry to penetrate the small voids
- Low gel strength: a gelling system restricts slurry movement
- No free water
- Appropriate fluid loss control
- Proper thickening time: to safely meet the anticipated job time

The following factors should be considered in designing the cement slurry for any squeeze operation:

a. Fluid Loss Control:

Fluid loss and filter cake growth rate vary directly i.e. higher the fluid loss; faster will be the filter cake build up. As such, while designing the slurry, fluid loss should be tailored to the formation type and the permeability so as to achieve a uniform cake build up against the squeeze interval.

The generally accepted API fluid loss rates are listed below:

- Extremely low permeability formation (less than 10 mD) 100 ml / 30 minute
- For High permeability formation, medium to high fluid loss slurry (200-500 ml/30 min) is initially pumped to allow fast filter cake development and diversion of slurry into smaller cracks.
- For voids and cracks, high fluid loss slurry is initially pumped to allow fast filter cake development, followed by low fluid loss slurry for squeeze.
- If the voids behind the casing are narrow channels or microfractures into which deep penetration is required, slurry with excellent fluid loss control (<50 ml/30 min) is required.

**CEMENTING OPERATIONS****b. Thickening Time:**

The temperatures encountered in squeezing can be higher than those of primary jobs, because fluid circulation before the job is usually less. For this reason, special API testing schedules/ software assisted simulation for squeeze cement slurry design should be followed to prevent premature setting. The additional feature of this API schedule is that it simulates the actual temperature the slurry is subjected to, when held near bottom for extended periods.

Thickening time shall be sufficient to assure slurry placement and reversing out of the excess slurry. For running squeeze method, requirement of thickening time should be less. Whereas for a hesitation squeeze method, higher pumping time shall be designed so that cement slurry remains in fluid stage till squeeze pressure is achieved.

c. Compressive Strength:

High compressive strength though desirable is not a primary concern for squeeze slurry design, as a partially dehydrated cement cake of any normal cement slurry will develop sufficient compressive strength.

10.2.3 Slurry volume

An optimum amount of cement is required to seal the void. The volume of slurry needed is generally inversely proportional to the injection pressure and directly proportional to the injection rate. The appropriate volume of cement slurry depends upon the length of the interval to be cemented and the placement technique to be used. A low-pressure squeeze may require only enough slurry to build a certain filter cake in each perforation tunnel. In many cases less than a barrel is sufficient. However, for job convenience and because of problems in placing the cement into the correct place to provide a seal, a 5-15 bbl batch should be prepared. A high-pressure squeeze, in which the formation is fractured, requires more volume of slurry.

10.2.4 Squeeze pressure

Squeeze pressure is the pressure at the injection point. In most cases, if the cement can be placed at the proper point, a successful squeeze can be obtained with 500 to 1000 psi standing pressure above the injection pressure. The pressure should be held for 10 to 15 minutes with no flow back. A safety factor of about 300 psi below formation fracturing pressure is reasonable for low pressure squeeze.

a. Low-pressure squeeze

The aim of this operation is to fill the perforation cavities and interconnected voids with dehydrated cement. The bottom hole treating pressure should be maintained below the formation fracturing pressure. The volume of cement is usually small, as no slurry is actually pumped into the formation. In low-pressure squeezes, it is essential that perforations and channels be clear of mud or other solids. If the well has been producing, such openings may already be free of obstructions. However, for newly completed wells, it may be necessary to clean the perforations before performing the squeeze job. A low-pressure squeeze should be run whenever possible as this technique has a higher success rate.

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b. High-pressure squeeze

In some cases, low-pressure squeeze of the perforations will not accomplish the objective of the job. The channels behind the casing might not be directly connected to the perforations. Small cracks or micro annuli that allow flow of gas may not allow the passage of cement slurry. In such cases, these channels shall be enlarged to accept a viscous solids-carrying fluid. In addition, many low-pressure operations cannot be performed if it is impossible to remove plugging fluids, or debris, from ahead of the cement slurry or inside the perforations. Placement of the cement slurry behind the casing is accomplished by breaking down the formation at or close to the perforations. Fluids ahead of the slurry are displaced in the fractures, allowing the slurry to fill the desired spaces. Further application of pressure dehydrates the slurry against the formation walls, leaving all channels (from fractures to perforations) filled with cement cake. However, during a high-pressure squeeze, the location and orientation of the created fracture cannot be controlled. A properly performed high pressure squeeze should have the cement as close to the wellbore as possible.

After a squeeze is obtained, the pressure should be bled off and the volume of returned fluid measured. The squeeze should then be re-pressurized, and the volume measured again. If the volumes are equal, this indicates that the squeeze has held, and the volume of fluid pumped has compensated for tubular expansion.

10.2.5 Slurry preparation

In most squeeze jobs, the amount of slurry involved is quite small, but the requirements of its quality are quite high, therefore special care shall be taken while preparing it. Use of a recirculating mixer or batch mixer is strongly recommended to ensure that the properties of the slurry pumped in the well are as close as possible to the slurry designed in the laboratory.

10.2.6 Evaluation of squeeze job

Pressure testing is the most common means of evaluating the success of the squeeze operation. Both positive and negative tests should be carried out. A squeeze job may appear successful when pressure is applied to the well bore but may fail to hold back the pressure from the zone into the casing.

a. Positive pressure test

After the WOC, dress the cement and test the casing integrity by applying required surface pressure. The pressure applied at the face of the perforation is predetermined at the job design stage. It may be the reservoir pressure or pressure equal to future working pressure in the well during fracturing or acidizing treatments but should not exceed the formation fracturing pressure.

b. Negative pressure test

A negative test or differential pressure testing of the well-bore may be done either by swabbing / compressor application to lower the fluid level or by displacing work over fluid with some lighter fluid. Negative pressure test should be conducted using pressure not greater than the expected maximum drawdown in the well when it is put on production.

When the objective of the squeeze is to repair a primary cement job, cement bond log (CBL/VDL) should be run to evaluate the effectiveness of the repair by comparing pre and post squeeze logs.

**CEMENTING OPERATIONS****10.2.7 Squeeze cementing procedure****a. Low pressure / High pressure squeeze cementing**

- i. Consult a CBL / VDL log prior to squeeze job.
- ii. Decide the point of perforation and perforate against a permeable formation at least 6 to 8 shots per foot for achieving better intake.
- iii. Carry out injectivity test against water cushion. If injectivity is found to be poor, acid job and back surging should be done to improve injectivity.
- iv. Follow all the recommended practices as given for a normal cement plug job so as to spot the slurry against the perforated interval.
- v. Then pull-out drill string sufficiently above the cement top. It is suggested to wash out excess cement, if any.
- vi. Close BOP and apply pressure through drill string to squeeze cement.
- vii. In case of low-pressure squeeze, maintain the bottom hole treating pressure below the formation fracturing pressure to fill the perforation cavities and interconnected voids with dehydrated cement.
- viii. For high-pressure squeeze, placement of the cement slurry behind the casing is accomplished by breaking down the formation at or close to the perforations by application of the bottom hole injection pressure exceeding the formation fracturing pressure.
- ix. Squeeze calculated volume of slurry and close the well under pressure for a period equal to slurry thickening time.

b. Block / circulation squeeze cementing

- i. For block squeeze, 2 sets of perforations are established, i.e. above and below the cement retainer.
- ii. Establish circulation through cement retainer behind casing with water or clean fluid to ensure good clean-up of the channels.
- iii. Maintain the downhole treating pressure below the formation fracture pressure when carrying out injectivity test or establishing circulation behind casing.
- iv. Calculate slurry volume keeping in consideration the annular volume and the volume below cement retainer which is to be filled with slurry.
- v. Use spacers ahead and behind cement slurry minimum length of 50 to 75 m to cover minimum 50 m of annuli, to avoid contamination.

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- vi. During displacement, monitor free falling /U tubing of cement slurry by controlling through choke. Displace cement up to the tip of cement retainer so as to keep the cement inside the string and engage tubing string to retainer, and squeeze to fill cement slurry between the two perforations.
- vii. Disengage the string from retainer and pull out the string well above the top of upper perforations. Reverse wash and squeeze cement in the upper perforation (optional) and keep the well under final squeeze pressure.

c. Water /Gas shut off squeeze cementing

- i. For elimination of water intrusion or reduction of gas oil ratio, this squeeze cementing is carried out to seal all the perforations and then re-perforate a selected interval.
- ii. All procedures of low-pressure squeeze cementing should be followed for placement of cement slurry against the perforated interval.
- iii. In case of good injectivity, squeeze calculated volume of slurry through the perforations leaving a cement plug inside the casing. Squeezing should be done by hesitation method, so that final squeeze pressure is achieved.
- iv. In case of no injectivity, squeeze cement slurry at the maximum permissible squeeze pressure and close the well under squeeze pressure for a period equal to slurry thickening time.

11.0 CEMENTATION OF TYPICAL WELLS**11.1 CEMENTING DEEP WATER WELLS**

Cementation of surface and conductor casing are critical in deepwater wells due to the following main reasons:

- i. Where conductor pipes are cemented at less than 600 m; the formations below mudline are relatively young geologically and are not consolidated. These shallow formations have the potential for abnormally pressured saltwater sands, also known as shallow water flow zones. These over-pressured zones are prone to uncontrolled flows of massive amounts of salt water, especially after cementing operations. Uncontrolled flows from such sands have led to total loss of the well in several cases. Weak formations and pressured sands present very narrow margins between pore pressure and the fracture gradient, which can cause a loss of cement returns and hence require lightweight cement slurries.
- ii. The low temperatures that occur at deep-water depths slow the cement hydration process and pose serious challenges for optimum strength development of cement slurry within a reasonable short period of time. This is particularly important from the point of view of WOC period because of the high day rates for deepwater rigs.

Thus, special attention to cement system design is required for cementing of conductor and surface casing in deep water areas. API Recommended Practice on Testing of Deepwater Well Cement Formulations, API RP 10B-3 or ISO 10426-3: 2019 Petroleum and natural gas industries- Cements and materials for well cementing- Part 3: Testing of deepwater well cement formulations or other equivalent standard should be referred to.

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A number of 'best practices' have been developed from lessons learnt while cementing in the deepwater, shallow water flow environment such as API Recommended Practice 65-1, Second Edition, June 2018 on Cementing Shallow Water Flow Zones in Deep Water Wells. The application of these practices will aid in successfully cementing the casing in deep-water conditions.

11.2 CEMENTING HIGH ANGLE / HORIZONTAL WELLS

Effective well completion is the key to obtaining the potential benefits of high angle / horizontal wells. Successful primary cementing is one of the vital components for effective completion in such wells.

Though the horizontal sections of the horizontal wells are mostly completed without cementation yet the achievement of good quality cementation in the previous intermediate casing, which is frequently highly deviated, is very critical for zonal isolation. Cement slurry properties thus have significant role in achieving zonal isolation for high angle well bore cementation and therefore need proper attention in the design stage.

11.2.1 Cement slurry design for high angle / horizontal wells

For designing cement slurry for horizontal/ highly deviated wells, proper estimation of BHCT is important. As the change in true vertical depth is significantly less with increasing measured depth in highly deviated wells, the conventional API table cannot be used for estimating the BHCT from BHST. Temperature simulating software programs should be used to predict, with a degree of accuracy, the accurate BHCT for use in slurry design of high angle wells. By using cement job simulator, as a design tool in the pre slurry design stage, the effective slurry density for safe placement during the job can also be predicted. The two most important key factors to be considered for deviated/horizontal well bore cementing are slurry stability and fluid loss.

i. Slurry stability

Good displacement practices minimize the occurrence of inter zonal communication channels that are created when cement bypasses the mud during primary cementing. However, communication channels can also be created after the cement has been displaced but while it is still in fluid state. These channels arise from cement slurries that are unstable and cause sedimentation and production of large quantities of free fluid when static. Though an unstable slurry may also cause zonal isolation problem in vertical wells, but it will be particularly harmful in a deviated or horizontal well where free fluid can collect along the high side of the annulus and sedimentation can result in a highly porous, low strength & low-density channel, which in turn may contribute to zonal communication and gas migration. Therefore, stable slurry should be designed to minimize the risk of annular communication and gas migration. Slurry stability is dependent on two properties i.e. free water and sedimentation. API Recommended Practice 10B-2 on "Recommended Practice for Testing Well Cements" specifies requirements and gives recommendations for the testing of cement slurries and related materials under simulated well conditions. Equivalent standard like ISO 10426-2 can also be referred.

Free water should be maintained at zero. Measurement of free water percentage at ambient conditions has been found to be inadequate for horizontal/high angle well conditions. More realistic procedure for free fluid determination at simulated bottom hole conditions and well-bore deviation as given in API RP 10B-2 or equivalent standard should be followed.

ii. Fluid loss

Fluid loss control is particularly important in high angle / horizontal wells, because slurry exposure to long permeable sections is more extensive than in vertical wells. Loss of mix water from the slurry, during displacement will reduce the slurry's thickening time and increase its viscosity. As the slurry loses water it forms a filter cake against the permeable formation wall. If uncontrolled, it changes the flow regime of the slurry from the calculated one to some unpredictable value and may result in excessive ECD. Low

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fluid loss rates are necessary to preserve the carefully designed rheological properties of the slurry. To reduce the amount of fluid loss from the slurry to the permeable reservoir section, API fluid loss of less than 50ml/ 30 minute tested with a differential pressure of 1000 psi and at bottom hole temperature should be maintained as recommended in API RP10B-2 or equivalent standard.

11.3 CEMENTING COAL BED METHANE (CBM) WELLS

The critical difference between gas bearing coal seams and conventional oil and gas sandstone reservoirs is the cleat system of the coal. Cleats are a network of natural fractures within the coal, which facilitates gas production in CBM wells. The cleats not only act as the drainage path for the gas during exploitation but also are conduits for undesired cement slurry filtrate loss / whole cement loss during cementation operations. Loss of cement filtrate and especially whole cement into the cleat system causes productivity damage. Since most CBM wells produce gas at rates far below conventional gas wells, this loss of gas production can seriously affect the economics of the venture. Over and above this factor, the requirement of long cement column rise because of completion of multiple coal seams in the same well and the need for very good zonal isolation for multiple hydro-fracturing treatments, to increase the permeability of coal seam, makes the cementation even more critical.

Cementing casing in wellbores that have penetrated coal-beds therefore requires special considerations because of the above circumstances. The cement slurry should be designed to accommodate the following conditions:

- i. Low bottomhole pressure, low breakdown pressure
- ii. Requirement of long cement column
- iii. Hole enlargement in the vicinity of the coal
- iv. Natural fractures in the coal
- v. Potential for low pressure gas leakage
- vi. Potential for cement channelling
- vii. Pressure near casing burst pressure during hydro-fracturing job
- viii. Multiple zones for completion and stimulation

Most coal well completions are simply a production string set through a shallow surface casing. A few of the deeper completions utilize a drilling liner to depth just above the major coal, then a short liner through the coal formation. In both circumstances, the surface casings are generally 60-90 m of 9-5/8" casing set in a 12-1/4" hole.

In consideration of higher compressive strength requirement for hydro-fracturing job, the use of normal weight cement slurry in CBM wells is ruled out as in that case the ECD during the cementation job will be more than the fracture opening pressure of the coal seams resulting in losses during cementation. Moreover, as the cleat network in the coal seams provides the permeability for fluid flow in coal, use of normal weight slurry may damage the formation due to cement invasion inside the cleats.

In consideration of all the above factors, the cementation in CBM wells should be carried out either in stages by use of stage collars or by using lightweight slurries with good compressive strength.

The production from a CBM well mainly depends on a properly executed hydro fracturing operation to increase the permeability by interconnecting the cleats in the coal seams. The success of a fracture job

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highly depends on the degree of isolation achieved in the annulus by cementation. Considering this importance of cementation jobs in a CBM well, the use of a cement job simulator as design tool is recommended. The simulator can generate critical information that provides the most efficient design of surface flow rates, thereby allowing maximum mud removal. By using the simulator prior to actually conducting the job, possible problems such as lost circulation or fluid invasion should be identified, and the placement schedule should be modified accordingly.

11.3.1 Slurry Design for CBM wells

The above listed cementing conditions are generally satisfied by one of the four slurry designs that have the necessary properties to combat typical problems in CBM wells i.e.

- Pozzolon mix cement blend with small additions of bentonite.
- Silicate based cement.
- Foam cement systems; and
- High performance lightweight slurries developed based on particle packing technology.

The tight economic constraint of coal well completions is also accommodated when using these slurry designs.

A 50/50 blend of pozzolon mix cement with 4% bentonite is generally used at a slurry weight of 12.7 to 12.8 ppg in the lead cement and the same slurry mixed at 13.5ppg, or occasionally neat cement mixed at 15.6ppg is used as tail slurry. Low fluid loss additive that does not delay thickening time and do not cause increased slurry viscosity should be used in areas where the leak-off is high, and lost circulation materials such as gilsonite are often included to help prevent cement contamination of the fractured coal.

In areas with zones that will not support cement weight of 12 to 14 ppg, silicate-based cement system should be used. These slurries exhibit excellent fluid loss characteristics, low slurry viscosity, essentially no free water, and high early compressive strengths. The normal slurry weights of silicate-based cement are from 11 to 13 ppg (1.32 to 1.56 sp. gr.).

The third option for CBM wells is foam cement. Foam cement should be used when the well bore integrity requires slurry weight under 11 ppg. Nitrogen foam cement has a density approaching that of water while still providing adequate compressive strength. Ductile foam cement is also resistant to fracturing when perforating. A great number of shallow, low-pressure wells are cemented with the foam cement system. Foam cement usually is formulated with basic cement, foaming agents, stabilizing agents, and nitrogen. This slurry would be most economical except that many of the coal wells are located in areas that do not have nearby nitrogen facilities, so the cost goes up. Relative cost and probability of success when using foam cements is usually compared to the alternative of using a stage collar with other cement formulations.

The fourth option that is very appropriate for application in CBM wells is the recently developed high performance lightweight slurries based on particle packing technology. The density range of these slurries is from 9.5 to 12ppg (1.14 to 1.44 sp. gr.). These slurries in spite of their low weight, develop high compressive strength and fulfil the cementing requirement of CBM wells i.e. lightweight to prevent losses and cleat damage during cementing as well as adequate set cement strength for hydro-fracturing.

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Regardless of the slurry chosen to cement CBM wells, a flush should be used for conditioning the hole to enhance the quality of bond of the cement and to limit the penetration of cement into the fractures of the coal. Flushes are especially suited for removing coal fines and uncirculated drilling debris from the annulus to reduce bridging of these materials in the annulus ahead of cement.

12.0 WELL ABANDONMENT

Proper well abandonment shall prevent pressure build-up, or cross flow, in the well and its surroundings by isolating all permeable hydrocarbon zones and water zones of different pressure regimes from each other. Proper abandonment shall also prevent contamination of freshwater aquifers, prevent leakage of any wellbore fluids to the surface, and minimize the consequential effects of well operations on the environment. The abandonment procedure given in this section applies to a well

- a. That has completed its life and is a non-productive well.
- b. That has ceased to produce oil or natural gas.
- c. That is no longer operational
- d. Where operations are temporarily abandoned

If Operations in the well are temporarily abandoned due to any operational / regulatory or any other reason and there is a plan to reactivate the well in future, procedure described under temporary Abandonment under **section 12.2** below should be adopted.

A. Planning & Preparation for abandonment

Prior to preparation of the abandonment programme the following shall be established between the appropriate departments:

- a. Identify permeable zones and identify those that contain hydrocarbons.
- b. Ascertain the potential maximum formation pressure.
- c. Identify zones that are capable of cross flowing.
- d. Identify zones of poor cement bond in the annuli and potential for annular pressure buildup.

An abandonment plan shall be prepared and approved by the multi-disciplinary team. Well barriers shall be installed which will include all annuli extending to the full cross section of the well and seal both vertically and horizontally, depending on if there is plan for re-entry or not.

B. Considerations for well abandonment operations planning:

- i. If there are porous zones not covered during the primary cementing of intermediate casing, remedial cementing should be carried out to isolate these intervals, especially where hydrocarbons are present. This is necessary to prevent hydrocarbon migration through the annular cement and to avoid pressure buildup in the annulus. Cement behind casing around hydrocarbon bearing or high-pressure water bearing zones should have good cement bond (both formations to cement and cement to casing) for 5 to 10 meters above and below the intervals.
- ii. It shall be ensured from cementing operation records (volume of cement slurry pumped, returns during cementing, cement plug top tagged inside casing) that surface casing is cemented up to surface and freshwater zones behind it are isolated.

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- iii. Well and the annuli of the well shall be stabilized before abandonment process.
- iv. While placing abandonment plugs/barriers, it should be ensured that the wellbore including the space between the cement plugs is filled with fluid of sufficient specific gravity and other properties so as to enable it to withstand any subsequent pressure which may develop in the wellbore.
- v. A bridge plug or cementing retainer with minimum 30 meters cement plug on top of it will be accepted as a barrier.
- vi. A cemented shoe track shall be accepted as a barrier if the casing is not perforated and the shoe track is verified as per Clause 9.3.
- vii. Cement plugs shall be verified as per Clause 4.3.
- viii. If a cement plug does not meet the requirements, repeat job shall be performed on the top of the existing plug.

12.1 PERMANENT WELL ABANDONMENT

The process of permanent abandonment essentially consists of down hole abandon plug(s), intermediate plug, top abandon plug and/or surface plug.

12.1.1 Downhole abandonment**a. Downhole abandonment in open hole**

- i. In uncased portion of well, firm cement plug shall be set to extend from a minimum of 30 m below the bottom to 30 m above the top of any oil, gas, or freshwater zones.
- ii. For longer drilled intervals and multi porous zones in the open hole, where covering all porous zones is difficult with one cement plug, two or more cement plugs shall be placed. The lower most cement plug shall extend 30 meters below the bottom most porous zone (or till drilled depth) and the upper most plug shall extend 30 meters above the upper most porous zone.
- iii. In addition, minimum 60 meters of firm cement shall be set (minimum 30 m below and 30 m above the casing shoe). This cement plug will be back up to the plug placed against the hydrocarbon bearing zone(s).
- iv. In case there is no oil, gas, or water zone, the open hole should be isolated by placing one firm cement plug of minimum 60 meters across casing shoe (minimum 30 m below and 30 m above the casing shoe)
- v. For abandonment of wellbore with expected or known lost circulation conditions, Open Hole Gravel Pack (OHGP) completion installed or any other technical limitation, a mechanical barrier like bridge plug / cement retainer should be set as close to the casing shoe as feasible, with minimum of 30 m of cement on top of the device.

**CEMENTING OPERATIONS****b. Down hole abandonment in cased hole**

- i. If there are perforations in casing, cement slurry shall be squeezed into the perforation(s) during permanent abandonment. Firm Cement plug shall be of at least 60 m, (minimum 30 m below to 30 m above the perforated interval or down to a casing plug, as applicable). If multiple zones are present but does not require isolation from one another, isolation of all zones from the rest of the well can be achieved by squeezing & plugging the shallowest zone.
- ii. For known loss circulation situations, it would be preferable to provide a platform (like bridge plug or cement retainer) just above the loss zone for supporting the cement plug. Cement plug of minimum 30 m shall be placed over the platform to be an acceptable barrier.
- iii. When there is completion packer(s) in the hole that will not be retrieved, the perforated interval below the packers should be isolated by squeezing cement through the packer. A firm cement plug of minimum 30m should be available above the packer after squeeze, which will be tagged and pressure tested.

12.1.2 Intermediate abandonment plug

- i. An intermediate firm cement plug of minimum 60 m length shall be placed in the production casing at an intermediate depth. In case of liner, intermediate firm plug of minimum 60 m shall be placed keeping 30 m below and minimum 30 m above liner hanger top.
- ii. Casing patch, casing failure point, cement squeeze intervals shall be covered by firm cement plug up to 30 m below and 30 m above the extremities.
- iii. If casing(s) are cut and retrieved thereby leaving one or more stub inside the next larger casing, abandonment cement plug(s) shall be set so as to extend to minimum of 30 m inside the deepest stub and 30 m above the uppermost stub, covering all the annuli inside the outer casing.

12.1.3 Top abandonment

Top abandonment for isolating and securing the annulus:

- i. Check each annulus one by one for any activity and subdue if necessary.
- ii. Perforate all Casing Annuli simultaneously ~25m above Surface Casing Shoe depth. Establish circulation in each annulus one by one.
- iii. Pump a calculated volume of cement slurry in the innermost casing and try to raise minimum 30 m cement slurry in each annulus one by one by operating annulus valves and closing BOP such that minimum 30 m plug remains inside casing above perforation. Pump/Squeeze slurry up to 1,000 psi/ within LOT value.
- iv. **For Offshore Wells with Sub-surface BOP:** Place a firm cement plug of 100 m length such that top of cement plug is at about 200-300 m below well head. No surface plug is required as annuli are closed by energizing the sealing element. Corrosion cap to be installed.

**CEMENTING OPERATIONS****12.1.4 Surface Plug before capping of the well:****a. For Onshore Wells:**

- i. Outer most annulus to be top filled with cement slurry through tubing after removal of Well Head if cement could not be raised in this annulus.
- ii. If casings are retrieved, place a progressive surface plug of minimum 60 m height covering all the annulus and 30 m inside inner most casing.
- iii. After removing well head, a plate (Minimum thickness of 10 mm) to be welded on the outermost casing with vent line and provision for pressure gauge.
- iv. Marking of the abandoned well shall be done as per requirement.

b. For Offshore Wells with Surface BOP:

- i. Check for any annulus activity and subdue if any activity is observed.
- ii. Cut and retrieve all casings or open and retrieve casings from MLS if used.
- iii. Place a balanced plug of at least 30 m inside inner most casing and raise up to 4 to 5 m below the mud line, covering all casing stubs.
- iv. Cut and retrieve conductor casing from about 2-3 m below seabed.
- v. Place Corrosion / (T/A) Cap which covers all the annulus and inner casing stub.

12.2 TEMPORARY ABANDONMENT

The temporary abandonment shall be carried out such that the well can be re-entered safely. The well is secured using pressure control equipment without compromising the barriers in place.

Wells that are temporarily abandoned should undergo regular monitoring and risk assessments to ensure their integrity.

For temporary abandoned subsea wells without monitoring, a program for visual observation shall be established. The frequency of visual inspection shall be established after risk assessment and should not normally exceed one year.

Procedures to be adopted for Temporary Abandonment:

- i. For wells with uncased open hole section, high viscous pill should be placed covering all permeable zones which are hydrocarbon bearing or over pressured water bearing. A cement plug of 60 m should be placed above the topmost permeable zone.
- ii. A firm cement plug of minimum 60 m length shall be placed across the innermost casing shoe (minimum 30 m inside casing and 30 m in open hole).
- iii. For wells with perforations in casing, a high viscous pill should be placed against the perforations. A firm cement plug of minimum 60 m shall be placed around 100 m above top perforations if re-activation of the well through the existing perforations is planned.
- iv. Where there is possibility of cement plug slippage down hole due to cement slurry density or losses, well activity or high gas content in formation fluid, cement plug should be set on a bridge plug or cement retainer as a suitable base (which will be set close to the casing shoe for isolating open hole section or just above top perforations as the case may be).

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- v. An intermediate cement plugs shall be placed close to the potential source of inflow, covering all possible leak paths. In case of liner, intermediate firm plug of minimum 60 m shall be placed keeping minimum 30 m above liner hanger top and 30 m inside liner.
- vi. A top abandonment plug of minimum 60 m shall be placed at around 300 m to 500 m below surface/ mud line.
- vii. For temporary abandonment of Onshore wells and offshore wells drilled by jack up rig on platform, the well should be secured with a X-Mas tree or other means with a kill string lowered.
- viii. For temporary abandonment of offshore wells drilled by jack up rigs without platform, dual seal type T/A caps should be installed on MLS (Mud Line Suspension System). A single temporary abandonment cap shall cover both production bore as well as annulus.
- ix. For temporary abandonment of Sub Sea well, corrosion cap shall be installed.
- x. For temporary abandonment of Subsea wells with Lower completion Installed:
 - a. Wells with lower completion installed may be temporarily abandoned with two mechanical barriers, both barriers shall be individually verified after setting. The lower mechanical barrier shall be set above the uppermost formations but as close to the perforation as feasible. Further, the lower barrier shall be verified with both positive & inflow test.
 - b. Visual inspection and risk assessment should be carried out at least once a year.

13.0 CEMENTING EQUIPMENT PRACTICES

Oil well cementation is an important and critical operation, which is accomplished at site in a relatively short period of time. Established safety procedures should be followed to avoid any untoward accident during cementation. To achieve this, pre-job safety meeting should be carried out with the rig crew. Following are some distinctive features of cementing operations.

- i. Simultaneous running of various equipment requiring co-ordination
- ii. Several liquid chemicals and additives used in cementation.
- iii. High pressure fluids and air involved in cementation.
- iv. Simultaneous presence of crew of different disciplines.
- v. Special attention during cementation during the nighttime.

The cementing in charge should ensure that all cementing service personnel and rig crew follow the practices outlined below:

13.1 PRE-DEPARTURE CHECKS OF MOBILE CEMENTING EQUIPMENT

- i. Oil level, HSD level, radiator water level, steering hydraulic oil level, battery connections, tyre pressure etc. should be checked.
- ii. After initial warm up of engine, check engine oil pressure, water temperature, air pressure, brake application, auto electrical light indications etc.
- iii. Fire extinguisher, spark arrestor in engine's exhaust pipes and a first-aid kit should be available in all cementing vehicles.

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- iv. Accessories like high pressure lines, valves, swivels, jet mixers, rubber hoses etc. should be clamped and fastened to avoid any loss and third-party injury while plying the cementing vehicle on road.
- v. While lifting the cabin of cementing vehicle for chassis & engine check-up/ repair, no person should be allowed to stand underneath till the cabin is locked and properly clamped in position.
- vi. All vehicles should be equipped with reverse audio-visual horn in working condition.

13.2 MOVEMENT & PLACEMENT OF CEMENT VEHICLE, HOOKING UP OF CEMENTING LINES AND PRE-JOB PLANNING

- i. Onshore cementing operator should be conversant with traffic signals and road safety rules and should have valid licence to drive heavy vehicles. Mobile vehicles should have valid road fitness.
- ii. Before moving the cementing units/ mobile silos to new well sites, route survey for road condition, sharp bends on road, bridge capacity for type & weight of vehicle, bridge clearance at top and safe clearance from live electrical wires on top should be carried out.
- iii. While carrying out any maintenance work beneath chassis, the electrical cut out switch at battery should be put in off position and starter key should be kept in safe custody.
- iv. While reversing a cementing vehicle, the driver should be certain that the sides and backing area are clear. The vehicle should not be reversed without a guide/ banksman.
- v. The use of appropriate Personal Protective Equipment (PPE) as specified in OISD-STD-155 shall be strictly enforced.
- vi. Personnel shall not wear metal watch straps and jewelry or other ornaments subject to snagging or hanging and causing injury while on the worksite.
- vii. A pre-job meeting should be held to ensure proper job layout and placement of cementing equipment.
- viii. Mobile cementing equipment positioning should be such that quick evacuation from the site is possible in case of an emergency. All vehicles should be placed with cabin facing away from the well and wooden wedge support should be placed behind wheels to minimise vibrations.
- ix. Place cementing pump / bunkers / mobile silos at least 1.5 - 2 m apart from each other and at least 25 m distance from the well head.
- x. Park all vehicles, which are not required for the job in safe areas away from the wellhead so as not to block the well site exits.
- xi. Wherever radioactive densometer is used, safe operating procedures and danger sign of exposure to radiation shall be prominently displayed. The sensor must be locked.
- xii. In hooking-up of high-pressure lines from cementing units, avoid crossing of two discharge lines. Lines should not be run under cementing trucks. Ensure proper anchoring of high-pressure lines to prevent accident in case of line failure.
- xiii. Use sufficient number of chocks to provide flexibility to discharge lines for reducing vibrations during cementing operations.
- xiv. Do not suspend discharge lines from cementing head without safety chains.

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- xv. Use high pressure fittings and steel pipes which have been inspected and are in good condition.
- xvi. Cementing heads, manifolds, valves and plugs should be inspected, cleaned and lubricated prior to hooking up. Always clean threads before making up cementing lines.
- xvii. The cementing in-charge shall inspect the hook up thoroughly prior to testing lines
- xviii. Care should be taken to avoid damage to the threaded pin end and stopper of cementing head during handling and tightening to the casing. Cementing head shall be secured to the links by safety chains. Thread protectors shall be used on all exposed male threads of circulating subs or cementing head to avoid thread damage.
- xix. Only steel lines should be used for releasing pressure and checking back flow from the well.
- xx. In electrical rigs, all cementing equipment should be earthed to the derrick structure to avoid electrical shock. Electrically powered cementing skid unit should also be earthed properly.
- xxi. NDT testing of cementing head, safety valves and high-pressure lines of cementing units should be done at an interval of 3 years.
- xxii. High pressure lines shall be tested with water at 1.5 times the maximum pressure expected in pumping operations.
 - a. Before testing, all persons shall be vacated from the vicinity of high-pressure lines. No one shall be permitted to step across, stand on or straddle or hammer on any pressurised line.
 - b. Do not allow anyone to take up line leakage repair operation until –
 - i. Cementing in-charge notifies the repair plan,
 - ii. Pressure has been released from the line,
 - iii. The release valve is left open during repairs and
 - iv. The flow has stopped from the bleed-off line.
- xxiii. During pre-job safety meeting, cementing in-charge should outline the job procedure, define pressure limits, discuss safety measures and give brief on emergency procedure and any unsafe conditions to all personnel participating in the job. He should describe the sequence, volumes and pumping rates. Also, duties of persons during cementing job including equipment operation, mixing of chemicals, operation of valves, bulk delivery, cementing head and maintenance should be fixed. He should also review communication system, which plays an important role in executing cementing job. Entire cementing operation should be controlled by cementing in-charge to avoid any confusion in executing sequence of operations during cementation.
- xxiv. In many offshore rigs, the cementing unit is placed in relatively congested closed space. The cementing in-charge should ensure that all exhaust fans are working and that all personnel present in the cementing room are wearing facemasks.

13.3 PUMPING JOB

- i. Safety announcement shall be made before pressurising the cementing lines.
- ii. All valves in discharge lines shall be checked to see that they are open before starting pumping.
- iii. No pumping shall take place while any personnel are working on, above or below rig floor level.
- iv. Flammable or combustible fluids shall not be placed in open displacement tanks on cementing equipment.

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- v. Pumping of acid with cementing unit should be avoided. In case it cannot be avoided, it should be ensured that all the valves, caps, lines etc. are fitted correctly and only required persons are present in the vicinity. After the job, the cementing unit and the lines should be washed thoroughly so as to remove the acid totally.
- vi. When pumping into any system, be sure to have an accurate pressure gauge. Start slowly with little throttle to confirm that system is open. Be alert for any closed valves.
- vii. A pressure-chart to continuously record pumping pressure should be available for all cementing jobs. Pressure chart should be supplemented with pumping sequence volume, time and rate.
- viii. The pumping pressure should not exceed the pressure rating of connections used such as unions, chocks, valves, crossovers etc.
- ix. When transferring or venting material through an open-ended hose, its end should be secured properly to prevent its whipping around.
- x. Loaded cement bunker or mobile cement silo should be parked on jacks at drill site.
- xi. During slurry mixing and chemical additions, adequate precautions should be taken to avoid chemical / additive contact with skin, eyes and clothing. Appropriate personal protective equipment like safety goggles, respirators, dust or vapour masks, face shield, rubber gloves, shoes and hearing protection should be worn. Fumes of defoamer should not be inhaled. Material safety data sheets of cement and its additives should be available at site
- xii. Review method and hazards thereof in handling, transferring and chemical mixing as well as proper mixing sequence.
- xiii. Proper illumination should be provided in the operational area especially at slurry mixing point and additive mixing system to ensure safe and effective job execution.

13.4 RIGGING DOWN

- i. After cement job is over, all pumps, lines and hoses should be flushed before rigging down. Valves and caps on all piping of each unit should be opened /removed to allow complete drainage of any fluid in the piping. Piping choked with cement slurry will damage cementing equipment and may lead to major breakdown.
- ii. Before dismantling the line, pressure must be released to zero. Tightening or loosening of connections under pressure shall not be done.
- iii. The air pressure in pneumatic bulk silo / mobile cement silo should be relieved before the vehicles are moved off location.



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- iv. Chicksans, high-pressure lines, valves, swivels, hoses with end connections should be safely shifted from rig floor to ground.

13.5 BULK CEMENT HANDLING PLANT

- i. All cement silos and other pressurised vessels shall be emptied and pressure tested as per standards. Safety valves and pressure gauges attached to each vessel should be checked for proper functioning.
- ii. Proper functioning of air dryer should be ensured to get rid of moisture in the air line of silos to prevent cement lump in the system and provide consistent dry cement supply for slurry mixing.
- iii. The discharge of air, dust and cement from vent line shall be in accordance with environment standards in force.
- iv. Personnel engaged in bulk handling plant operation shall use all personal protective equipment as per OISD-STD-155.

**CEMENTING OPERATIONS****14.0 REFERENCES:**

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- API Specification 10A / ISO 10426-1, Specification for Cements and Materials for Well Cementing, Twenty-fifth Edition, April 2022.
- API Recommended Practice 10B-3 / ISO 10426-3, Petroleum and natural gas industries-Cements and materials for well cementing. Recommended Practice on Testing of Deepwater Well Cement Formulations, First Edition, July 2020.
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- OISD-RP-238 (Well Integrity)
- OISD-RP-242 (Drilling and testing of HPHT wells)
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- Deepwater Cementing Challenges, K. Ravi, E.N. Biezen, S.C. Lightford, A. Hibbert and C. Greaves, SPE # 56534, SPE Annual Technical Conference, Texas, October 3-6, 1999.
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CEMENTING OPERATIONS

15.0 ABBREVIATIONS USED:

1. API - American Petroleum Institute
2. Bc- Bearden consistency
3. BOP- Blow Out Preventer
4. BHCT- Bottom Hole Circulating Temperature
5. BHST – Bottom Hole Static Temperature
6. BPM – Barrels Per Minute
7. CBL/VDL – Cement Bond Log/ Variable Density Log
8. CBM – Coal Bed Methane
9. CET – Cement Evaluation Tool
10. ECD – Equivalent circulation density
11. HP-- High pressure
12. HPHT – High Pressure, High Temperature
13. HSD – High Speed Diesel
14. LCM – Loss Circulation Material
15. LOT – Leak off Test
16. LWD – Logging While Drilling
17. m – Metre
18. mD – Milli Darcy
19. ml - Millilitre
20. MLS – Mud Line Suspension
21. m/sec – Metre Per Second
22. NDT – Non-Destructive Testing
23. NRV – Non Return Valve
24. OEM – Original Equipment Manufacturer
25. P&A – Plug and abandonment
26. PJSM-Pre Job Safety Meeting
27. psi – Pounds Per Square Inch
28. Pv – Plastic Viscosity
29. PPG – Pounds Per Gallon

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- 30. rpm – Revolutions Per Minute
- 31. SPE – Society of Petroleum Engineers
- 32. SPM – Strokes Per Minute
- 33. TA – Temporary abandonment
- 34. TOC – Top of Cement
- 35. USIT -Ultrasonic Imaging Tool of Schlumberger
- 36. UCA- Ultrasonic Cement Analyzer
- 37. W/Pr- Working Pressure
- 38. WOC-Waiting On Cement
- 39. Yp – Yield Point
- 40. RTTS packer- Retrievable Tubing Testing Squeeze packer



CEMENTING OPERATIONS

Annexure I

EQUIPMENT CHECKLIST

	Activity/Check	Yes/ O.K.	No/ Not O.K.	Remarks
1.	Well-site identification /location?			
2.	Route survey (Road condition, bends, high tension wires, bridge load limit etc.) done			
3.	Site ready for placement of the cementing unit & bunkers / silos?			
4.	Securing of all hoses, steel pipes, chocks, swivel, hopper etc. with chains/clamps on the mobile unit done			
5	Engine - Lube oil level & condition - Transmission oil level - Radiator water level - Radiator cap - Oil pressure gauges - Temperature gauge - Emergency shut down system - Any leakages from diesel/oil lines - Air filters - Oil filters - Condition of spark arrestors on engines			
6	Chassis - Brakes - Air pressure for brakes - Parking brakes - Condition of hoses and clamps - Tyre pressure - Condition of tyre treads - Availability & condition of jacks - Shroud/guard on rotating parts			

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7	Light and Power of cementing unit - Head lights - Dippers - Parking light - Turning indicator light - Reverse lights/horn - Brake light - Condition of battery - Availability of battery cables for jump starting - Battery cut-out switch			
8	Cabin - First aid box - Fire extinguisher			

HIGH PRESSURE CEMENTING PUMPS

	Activity/Check	Yes/ O.K	No/ Not O.K.	Remarks
1	Condition of plunger packing / liner-piston			
2.	Discharge through return line at low and high discharge rate (to check valves & seats) –smooth or intermittent			
3	Minimum two no of calibrated main discharge pressure gauges			
4	Lube oil level and condition			
5	Lube oil pump pressure			
6	Condition of heat exchangers for - any leakage (if any, plug leaking tube or replace heat exchanger) - cooling efficiency (clean tubes if cooling not sufficient)			
7	Barrel counter (Flow meter) working			
8	All high pressure valves of discharge and return manifold pressure tested and holding			
9	High pressure mixing manifold available for any eventuality			
10	Safety valve on discharge line fitted and its relief pressure set at maximum 10% above max. working pressure			
11	Visual inspection of suction header and lines for any debris or cement sheath (clear same, if any)			
12	Densometer (if installed) calibrated with water			

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MIXING SYSTEM

	Activity/Check	Yes/ O. K	No/ Not O.K.	Remarks
1	Mixing pump discharge pressure (as per OEM) If low, check - starving of suction (suction line partially clogged) - condition of gland packing/seals - adjust Impeller clearance - rpm of the pump (If all above are O.K., dismantle pump and replace worn casing and/or impeller)			
2.	Recirculating pump discharge pressure (as per OEM) If low, check - starving of suction (suction line partially clogged) - condition of gland packing/seals - adjust Impeller clearance - rpm of the pump (If all above are O.K., dismantle pump and replace worn casing and/or impeller)			
3	Check mixing jet and bowl for - correct slit/holes size for the slurry weight and mixing rate - cleanliness of bowl for proper suction - condition of O- ring on hopper to prevent suction of air - worn jet - vacuum killing cup working in case of RCM - knife gate movement and sealing of bowl when completely closed in case of PSM			
4.	Supply rate of mix fluid to the displacement tanks is sufficiently higher than mixing rate			
5.	Displacement tanks and mixing tub or batch mixer are properly cleaned			
6	Isolation valves of displacement tanks working			



CEMENTING OPERATIONS

DISCHARGE LINES

	Activity/Check	Yes/ O. K	No/ Not O.K.	Remarks
1.	All Discharge lines are pressure tested at 1.5 times maximum anticipated pressure during the job for 5 minutes			
2	Announcement over paging system and/or barricades put around pressurised lines for personnel to stay away from lines while pressurised			
3.	All lines including release line are secured with clamps and safety chains			
4	CWP (Cold Working Pressure) rating of steel pipes connected is higher than maximum anticipated line pressure			
5	In H ₂ S areas, pipes and fittings for sour gas service used			
6	All hammer unions are compatible and are of same pressure rating			
7.	Check for tightness of all hammer unions at zero pressure.			

BULK HANDLING SYSTEM

	Activity/Check	Yes/ O. K	No/ Not O.K.	Remarks
1	Supply of air for bulk transfer is dry			
2	Drained moisture from receiving tank			
3	Regulated pressure air supply to the Silos			
4	Mechanical integrity of silos/surge tank checked by closing all valves and charging to working pressure (30-40Psi)			
5	Relief valve on silos and surge tank working with relief pressure set within 10% of max. operating pressure			
6	Provision of proper dust collector in the vent of surge tank to maintain out coming air quality in accordance with applicable environment standards in force			

CEMENTING HEADS

	Activity/Check	Yes/ O. K	No/ Not O.K.	Remarks
	a. Surface cementing Head			
1	Head is designed to handle maximum pressure anticipated during the operation			
2	Condition of cap seal and x-over seal			
3	Plug holding shaft - retracting and protruding (check number of turns and position of shaft each time) - free movement of shaft - leakage from shaft seal			

CEMENTING OPERATIONS

	- locking device in place			
4	Free movement of plug release indicator & its lubrication			
5	One high pressure valve provided for connecting line to the head for single plug container and two high pressure valves for double plug container			
6	Swage (circulating head) with threads compatible with those of casing available			
	b. Cementing Head for sub sea operations:			
1	Inspect swivel of the connector to mandrel			
2	Condition of O-rings /seals of mandrel piston.			
3	Check threads of the connector to the top plug			
4	Check threads of plug connector and its shear groove. Note plug launch pressure			
5	Ball seal ring on bottom plug for free rotation			
6	Ball size and type for particular size of plug			
7	Note ball extrusion pressure			
8	Rabbiting (gauging) of all drill pipes above mandrel for the particular ball size done			

GENERAL SAFETY

	Activity/Check	Yes/ O. K	No/ Not O.K.	Remarks
1	MSDS (Material Safety Data Sheet) for different additives used in cementing available			
2	Eyewash available in working condition near mixing hoppers.			
3	Availability and use of prescribed personal protective equipment as per OISD-STD-155			



CEMENTING OPERATIONS

ANNEXURE II

MAJOR CHANGES IN REVISED EDITION

The following Major Changes were made.

1. **Introduction:** The standard's introduction has been updated to provide clearer guidance on cementing operations and to highlight the critical safety aspects involved.
2. **Definitions:** Most definitions like permanent abandonment, sustained casing pressure, gas tight slurry, thickening time, well integrity, leak off test etc. were introduced to clarify terminology relevant to onshore and offshore cementing operations.
3. **Plug verification:** The plug verification section was updated to reflect diverse testing scenarios, ensuring compliance with best safety practices.
4. **Stab in/ inner string cementing:** This section was modified for better clarity on stab in/ inner casing cementing.
5. **Casing integrity test/ Hermetical test:** A new sub section "Casing integrity test/ Hermetical test" was added in the standard.
6. **Planning and preparation for abandonment:** A new section "Planning and preparation for abandonment" was included in this version of standard.
7. **Top Abandonment:** Section on top abandonment was added in the standard.
8. **Surface Plug before capping the well:** The section was modified for better clarity for the surface plug.



CEMENTING OPERATIONS

Annexure III

FUNCTIONAL COMMITTEE

(First Edition - 1999)

NAME	ORGANISATION
LEADER	
Shri A.K. Hazarika	ONGC
MEMBERS	
Shri A.K. Mehra	ONGC
Shri N. K. Tyagi	ONGC
Shri T. H. Mazumdar	OIL
Shri D. K. Rajkhowa	OIL
MEMBER COORDINATOR	
Shri T.Singh	OISD

CEMENTING OPERATIONS

**Functional committee
(Second edition - 2008)**

NAME	ORGANISATION
LEADER	
D. SANYAL	ONGC
MEMBERS	
S.A. RAVI	ONGC
D.DASGUPTA	ONGC
PRADIP KUMAR	ONGC
BUDHIN GOSWAMI	OIL
S. CHANDA	OIL
MEMBER COORDINATOR	
U.K. ARORA	OISD